BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Data Analysis Report

Project ID: RP 25-26J-109

Project Title: Multimodal AI Framework for Early Parkinson’s Disease Detection

1. Introduction
   1.1. Background:

Parkinson's Disease (PD) is a complex neurodegenerative disorder characterized by the progressive loss of dopaminergic neurons, leading to a range of motor and non-motor symptoms. It is currently the fastest-growing neurological condition worldwide, affecting over 10 million people, creating an urgent need for accurate diagnostic tools. Traditional diagnostic methods primarily rely on clinical assessments and specialized imaging, which are often subjective and may only detect the disease after significant neuronal damage has already occurred.

Recent advancements in digital health technologies have opened new possibilities for early detection using digital biomarkers, such as changes in speech patterns, motor abnormalities, and cognitive decline. This research leverages a comprehensive multimodal approach that integrates speech analysis, wearable sensor data, cognitive-motor interaction assessments, and digitized drawing tests. By utilizing advanced machine learning algorithms and TinyML for on-device processing, the project aims to develop a robust, accessible, and privacy-preserving screening tool to identify PD in its earliest stages.

   1.2. Research Problem:

Current diagnostic frameworks for Parkinson's Disease face critical challenges, particularly in resource-limited settings where specialized imaging and expert clinical assessments are unavailable. Existing automated tools often focus on single modalities, which limits their sensitivity and generalizability across diverse populations. Consequently, many patients are diagnosed only after the onset of severe motor symptoms, missing the window for early intervention and personalized treatment strategies.

In this research, we address these gaps by developing an AI-driven multimodal diagnostic system that combines diverse data streams—including voice, motion, drawings, and cognitive tasks—into a unified prediction engine. The solution tackles the issues of accessibility and computational cost by employing TinyML-optimized models for real-time, on-device analysis, ensuring that the system is both non-invasive and scalable. Furthermore, to overcome the "black box" nature of traditional AI, this project integrates Explainable AI (XAI) to provide interpretable insights for clinicians, thereby enhancing trust and facilitating better patient management.

1.3. Objectives:

It Number	Objective	Objective number
IT22093332	To develop a privacy-preserving AI system for early Parkinson’s Disease detection by utilizing on-device TinyML powered by TensorFlow Lite for real-time feature extraction and classification. The methodology begins with gathering motion data from wearables using standardized protocols—emphasizing the inclusion of Sri Lankan cohorts for local relevance—followed by preprocessing raw signals to extract optimized time- and frequency-domain features directly on low-compute devices. These features drive the development and fine-tuning of machine learning models like SVM and LSTM using the PADS (Parkinson's Disease Smartwatch) dataset [1], which are then deployed within an integrated architecture featuring user interfaces for feedback and cloud storage for anonymized aggregates. The system is finally validated by evaluating performance metrics against clinical benchmarks such as UPDRS and assessing real-world usability to ensure robust, accurate screening.	1

IT22551498	To develop a mobile-compatible cognitive assessment component that models subtle cognitive changes associated with Parkinson's Disease and outputs a continuous cognitive risk score, enabling early-stage detection and seamless integration within a multimodal diagnostic system.	2
IT22079336	To develop an AI-powered speech analysis system for early detection of Parkinson's Disease (PD) by identifying subtle vocal biomarkers. The system will analyze both acoustic features (jitter, shimmer, pitch variability) and linguistic patterns (pause frequency, speech rhythm) using advanced signal processing and machine learning techniques. By creating an interpretable, mobile-compatible solution, this component will contribute to a comprehensive multimodal framework for scalable, accurate, and non-invasive PD detection.	3
IT22884060	To develop an AI-powered drawing analysis system for early detection of Parkinson's Disease (PD) by identifying subtle motor biomarkers from smartphone finger-drawing tests. The system analyzes both visual features (stroke patterns, drawing trajectory,	4

	line consistency) and kinematic patterns (velocity, acceleration, jerk, frequency-domain characteristics) using advanced deep learning and multi-modal fusion techniques. By implementing an interpretable, mobile-compatible Quad-Stream architecture that integrates spiral and wave drawing tasks, this component contributes to a comprehensive multimodal framework for scalable, accurate, and non-invasive PD detection	

2. Data Exploration
   2.1. Data Collection

Objective 1

The data for this research component was collected from a publicly available, peer-reviewed dataset related to wearable-based Parkinson's disease motion analysis. The selected dataset supports the identification and modelling of kinematic biomarkers associated with Parkinson's disease severity, specifically focusing on tremor and bradykinesia quantification.

The primary dataset was obtained from the PADS (Parkinson's Disease Smartwatch) Dataset [1], hosted on PhysioNet [2], which contains structured motion measurements collected during a cross-sectional prospective study at the University Hospital Münster. The dataset includes high-frequency sensor data from 469 participants, comprising individuals with Parkinson's disease (PD), differential diagnoses (DD), and healthy controls (HC) [1]. Motion data was captured using an Apple Watch Series 4 at a sampling rate of 100 Hz [1], which aligns perfectly with the acquisition frequency of the ESP32-based prototype used in this study.

The data collection protocol involved participants performing a series of standardized motor tasks—such as resting tremor assessment and hand rotation—under the guidance of a study nurse, as well as passive monitoring [1]. This approach ensured high ecological validity by capturing both clinical assessment metrics and subtle movement pathologies. The dataset was selected based on its high temporal resolution (100 Hz), extensive clinical annotations (including disease duration and diagnosis), and its suitability for validating TinyML inference models on edge devices .

Objective 2

The data used for this research component were obtained from the Parkinson's Progression Markers Initiative (PPMI), a large-scale, longitudinal, multi-center observational study designed to identify biomarkers of Parkinson's Disease (PD).

PPMI data were collected prospectively from multiple international clinical sites under standardized protocols. Participants include healthy controls, prodromal (high-risk) individuals, and clinically diagnosed Parkinson's Disease patients, allowing analysis across different disease stages. Data collection spans over a decade, enabling both cross-sectional and longitudinal analysis.

For this research component, only baseline clinical visit data were initially utilized to ensure consistency across participants and to avoid temporal leakage during model development. Baseline data represents the earliest available cognitive assessment for each subject, which is particularly relevant for early-stage and prodromal PD detection.

The cognitive data were collected using standardized neuropsychological assessments, administered by trained clinical personnel, ensuring high data reliability and clinical validity. These assessments measure multiple cognitive domains including processing speed, memory, attention, executive function, and visuospatial ability.

Demographic attributes such as age, sex, and years of education were also collected, as these factors are known to influence cognitive performance and are essential for fair and clinically meaningful modeling.

All data were obtained in de-identified form through the official PPMI data access portal, complying with ethical guidelines and data usage agreements.

Objective 3

The data for this research component was collected from publicly available, peer-reviewed datasets related to Parkinson's disease speech analysis. The selected datasets support the identification and modelling of speech-based biomarkers associated with Parkinson's disease severity and progression.

One dataset was obtained from the Oxford Parkinson's Disease Telemonitoring Dataset, which contains structured biomedical voice measurements collected during a six-month telemonitoring trial involving individuals with early-stage Parkinson's disease. Voice recordings were automatically captured in the participants' home environments using a remote monitoring system, enabling realistic and longitudinal data collection.

The second dataset (MDVR-KCL) consists of high-quality voice recordings collected in a controlled clinical environment at King's College London Hospital using a smartphone-based recording setup designed to simulate real-world phone call conditions. This approach ensured ecological validity while maintaining audio quality suitable for acoustic and linguistic analysis.

Both datasets are openly accessible for research purposes and were selected based on their relevance, clinical annotation quality, and suitability for machine learning-based Parkinson's disease analysis.

Objective 4

The primary dataset used in this study was obtained from a publicly available dataset published as part of a peer-reviewed study, Finger drawing on smartphone screens enables early Parkinson's disease detection through hybrid 1D-CNN and BiGRU deep learning architecture | PLOS One. The dataset includes anonymized time-series touch-motion data, collected while participants performed guided finger-drawing tasks, specifically spiral tracing and wave tracing, on a smartphone screen. During task execution, finger movements were recorded as continuous (x, y) coordinate trajectories over time.

The dataset comprises recordings from 58 participants, including 30 healthy controls and 28 individuals diagnosed with early-stage Parkinson's disease. It also contains relevant demographic and clinical information, such as age (56–77 years), sex, Hoehn–Yahr stage (1.0–2.5), and years since diagnosis for Parkinson's disease participants.

The smartphone-based finger-drawing dataset used in this study was obtained from a publicly available, peer-reviewed PLOS One publication and its accompanying supporting information files. The dataset is released under the Creative Commons Attribution (CC BY) license, which permits unrestricted use, distribution, and reproduction for academic and research purposes, provided that the original authors and source are appropriately cited. All data are fully anonymized, and no direct interaction with human participants was involved in this phase of the study.

To evaluate the generalizability of the proposed approach, external validation was performed using two well established benchmark datasets: PaHaW and NewHandPD. The PaHaW dataset was accessed after formally obtaining permission from the dataset custodians and signing the required legal and data usage agreement, while the NewHandPD dataset was obtained from its publicly available source [Welcome to the HandPD dataset home-page]. Both datasets contain digitized handwriting and drawing recordings collected using stylus-based input devices, providing heterogeneous acquisition conditions for cross-dataset evaluation.

As future work, and prior to the final viva, it is planned to collect new data from real participants using the smartphone-based data acquisition model developed in this study, implemented through the proposed mobile application. This data collection will be carried out under appropriate ethical approval and external supervision, with the objective of validating the proposed system in real-world clinical and community settings.

2.2. Dataset Description:

Data source	Description	Resource	Size	Key attributes
		Objective 1		
PADS - Parkinson's Disease Smartwatch Dataset	A comprehensive wearable sensor dataset collected from individuals with Parkinson's disease (PD), differential diagnoses, and healthy controls at the University Hospital Münster. It consists of high-frequency (100 Hz) motion data captured via an Apple Watch Series 4 during standardized clinical motor tasks (e.g., resting tremor, hand rotation) and passive monitoring.	Publicly available research dataset (PhysioNet)	469 participants (including 227 PD patients)	Participant ID, age, gender, diagnosis label (PD/DD/HC), disease duration, Levodopa medication status, MDS-UPDRS Part III scores, tri-axial accelerometer (X,Y,Z), tri-axial gyroscope (X,Y,Z).
		Objective 2		
PPMI (Parkinson's Progression Markers	A clinically curated dataset containing	PPMI Database	4,331 participants (17,252 records)	Cognitive test scores (e.g., MoCA, SDMT, HVLT, LNS),

Initiative) PPMI	standardized cognitive assessment scores and demographic information from healthy controls, prodromal individuals, and Parkinson's Disease patients. Baseline visit data were used to support early-stage disease analysis.			demographic variables (age, sex, education), cohort labels (Healthy, Prodromal, PD), visit identifiers
Objective 3				
Oxford Parkinson's Disease Telemonitoring Dataset	A tabular biomedical dataset consisting of speech-derived voice measurements collected from individuals diagnosed with early-stage Parkinson's disease during a six-month telemonitoring trial. Each record represents a single voice	Publicly available research dataset (Oxford University)	5,875 instances with 19 features	Subject ID, age, gender, time since baseline, 16 biomedical voice measures, motor UPDRS score, total UPDRS score

	recording captured automatically in the participant's home environment.			
King's College London Smartphone-Based Voice Recording Dataset	A clinical speech dataset comprising high-quality smartphone-recorded voice samples collected in a controlled hospital environment. The dataset includes scripted reading tasks and spontaneous speech, annotated with expert-assessed Parkinson's disease severity scores.	Publicly available clinical research dataset	Audio recordings from Parkinson's disease patients and healthy control participants (WAV format, 44.1 kHz, 16-bit)	Raw speech audio signals, health status label (PD/HC), Hoehn & Yahr stage, UPDRS II (Part 5) score, UPDRS III (Part 18) score
Objective 4				
Smartphone Finger-Drawing Parkinson's	Smartphone-based finger-drawing time-series data	Publicly available, peer-reviewed dataset	58 participants (28 Parkinson's	Time-series (x, y) touch coordinates, spiral and

Touch-Motion Dataset Finger drawing on smartphone screens enables early Parkinson's disease detection through hybrid 1D-CNN and BiGRU deep learning architecture | PLOS One)	collected from Parkinson's disease patients and healthy controls during spiral and wave tracing tasks, along with basic demographic and clinical information.	released as part of a PLOS One publication and accessed through the paper's supporting information files	disease patients, 30 healthy controls), with 116 drawing samples (58 spiral and 58 wave recordings)	wave drawing tasks, demographic information (age, sex), and clinical data for Parkinson's disease participants (Hoehn–Yahr stage, years since diagnosis).
NewHandPD Welcome to the HandPD dataset home-page]	Stylus-based handwriting and drawing time-series data collected from Parkinson's disease patients and healthy controls	Publicly available benchmark dataset obtained from the official dataset repository for academic research use.	66 participants (31 Parkinson's disease patients, 35 healthy controls), multiple drawing samples per subject.	Time-series pen trajectory data, spiral and handwriting tasks, kinematic features (position, velocity, acceleration), demographic information (age, sex).
PaHaW (Parkinson's Disease Handwriting)	Digitized handwriting and drawing recordings collected using stylus-based	Public benchmark dataset accessed after formally requesting	75 participants (Parkinson's disease patients and healthy	Stylus-based time-series trajectory data, spiral and handwriting

	input devices under controlled conditions	permission from the dataset custodians and signing the required legal and data usage agreement.	controls), multiple handwriting and drawing tasks	tasks, kinematic features, demographic and clinical information.

2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

	1	2	3	4
Data source 1 (PADS (Parkinson's Disease Smartwatch) Dataset, hosted on PhysioNet)	The PADS dataset is highly suitable for this research component as it provides quantitative motion-based features directly related to Parkinson's disease symptom severity. The dataset includes high-resolution tri-axial accelerometer and gyroscope data sampled at 100 Hz, which captures the			

	fine-grained kinematic details necessary to distinguish between pathological tremors and normal voluntary movements. These technical specifications align directly with the research objective of identifying subtle kinematic biomarkers associated with tremor and bradykinesia using low-power edge devices. Additionally, the inclusion of clinical metadata, specifically the MDS-UPDRS Part III scores, enables supervised learning approaches			

	for regression-based severity estimation and precise classification between PD patients and healthy controls. The dataset's unique combination of standardized clinical motor tasks (active tests) and passive monitoring enables the validation of the system under both controlled and "free-living" conditions. These characteristics make the dataset particularly appropriate for training and validating TinyML models focused on real-time, privacy-preserving			

	motion analysis.			
Data source 2		The PPMI cognitive dataset is highly suitable for the proposed research as it contains clinically validated cognitive assessment scores across healthy controls, prodromal individuals, and Parkinson's Disease patients. The availability of baseline cognitive measures enables the identification of early cognitive patterns associated with Parkinson's Disease, directly supporting the objective of designing a mobile-compatible cognitive screening module.		

		Additionally, the large sample size and standardized data collection protocols ensure robustness, reliability, and scalability for machine learning–based analysis.		
Data source 3 (The Oxford Parkinson's Disease Telemonitoring Dataset)			This Dataset is highly suitable for this research component as it provides quantitative biomedical voice features directly related to Parkinson's disease symptom severity. The dataset includes 16 clinically validated acoustic measurements, such as jitter, shimmer, and frequency-	

			based features, which align directly with the objective of identifying subtle vocal biomarkers associated with Parkinson’s disease. Additionally, the availability of motor and total UPDRS scores enables supervised learning approaches for regression-based severity estimation and early-stage disease analysis. The longitudinal nature of the data, with multiple recordings per participant collected in real-world home	

			environments, supports the development of a scalable and non-invasive monitoring framework. These characteristics make the dataset particularly appropriate for training and validating machine learning models focused on acoustic feature analysis.	
Data source 4 (MDVR -KCL Dataset)			This dataset is well suited for analyzing both acoustic and linguistic speech characteristics relevant to early Parkinson's disease detection. The dataset	

			contains high-quality raw audio recordings collected under control yet realistic phone-call conditions, making it ideal for extracting temporal, prosodic, and linguistic features, including pause frequency, speech rhythm, and fluency. The inclusion of expert-annotated clinical labels, such as Hoehn & Yahr stages and UPDRS scores, allows for supervised machine learning and classification tasks. Furthermore, the	

			smartphone-based recording setup aligns with the objective of developing a mobile-compatible and deployable solution, supporting real-world applicability and scalability. This dataset therefore complements the Oxford dataset by enabling end-to-end speech signal processing and linguistic pattern analysis.	
Data source 5 Smartphone Finger-Drawing Parkinson's Touch-Motion Dataset				This dataset aligns very strongly with the research problem and objectives, as the project focuses on early Parkinson's

				disease detection using smartphone-based finger-drawing data. It contains time-series (x, y) touch-coordinate data collected during spiral and wave drawing tasks, which directly support the objective of analyzing drawing-based motor impairments using a smartphone application. As the data acquisition method matches the proposed system, this dataset serves as the primary dataset for the study
Data source 6				The NewHandPD dataset aligns with the research objectives by

				providing stylus-based handwriting and drawing time-series data from Parkinson's disease patients and healthy controls. Although the data are collected using a different input device, the dataset enables external validation of the proposed approach. It helps assess whether the motor patterns identified from smartphone-based data remain consistent across different hardware and acquisition settings
Data source 7				The PaHaW dataset supports the research

				objectives as a widely used benchmark dataset for Parkinson's disease handwriting analysis. Its inclusion allows for comparative evaluation and robustness testing of the proposed methodology across heterogeneous datasets, strengthening the reliability and generalizability of the project outcomes

3. Methodology

3.1. Data Preprocessing:

Objective 1

	Transformation technique							
	Data Cleaning	Data Transformation	Feature Engineering	Feature Selection / Dimensionality Reduction	Outlier Detection and Handling	Data Scaling (Standardization)	Data Encoding	Data Cleaning
Data source 1	Initial 50 samples (~0.5 seconds) were trimmed from each recording to remove transient startup noise. Static/resting windows were removed by filtering windows where Acc_RMS < 0.05. Patient metadata was matched via file_list.c	Bandpass Filtering: Butterworth 4th-order filter applied (0.5–20 Hz) to remove gravity (DC offset) and high-frequency noise. Windowing: Data segmented into 256-sample windows (2.56s at 100Hz) with 50% overlap (128 sample stride). FFT: Mean-centered magnitude	16 features extracted per window: Accelerometer (8): RMS, Std Dev, Max, Min, Dominant Frequency, Tremor Power (3-7Hz), Tremor Ratio, Spectral Entropy. Gyroscope (8): RMS, Std Dev, Max, Min, Dominant Frequency, Tremor Power (3-7Hz), Tremor Ratio, Spectral	PCA (2 components) applied during EDA for visualization of class separability. t-SNE (2 components, perplexity =30) used for non-linear dimensionality reduction. All 16 features were retained for model training as they capture clinically relevant tremor	Windows with very low energy (Acc_RMS < 0.05) were removed to exclude resting/static periods that don't contain meaningful tremor information. This energy-based filtering ensured only active movement windows were	StandardScaler (Z-score normalization) applied:	Initial 50 samples (~0.5 seconds) were trimmed from each recording to remove transient startup noise. Static/resting windows were removed by filtering windows where Acc_RMS < 0.05. Patient metadata was matched via file_list.c	Bandpass Filtering: Butterworth 4th-order filter applied (0.5–20 Hz) to remove gravity (DC offset) and high-frequency noise. Windowing: Data segmented into 256-sample windows (2.56s at 100Hz)

	sv to ensure no missing labels.	signals converted to frequency domain using np.fft.rfft() .	Entropy. Magnitude calculation : sqrt(X^2 + Y^2 + Z^2) for rotation-invariant representation.	characteristics.	included in training.		sv to ensure no missing labels.	with 50% overlap (128 sample stride). FFT: Mean-centered magnitude signals converted to frequency domain using np.fft.rfft().

Objective 2

	Transformation technique					
	Data Cleaning & Filtering	Data Transformation & Encoding	Feature Engineering & Selection	Handling Missing Data	Data Scaling & Normalization	Outlier Detection & Handling
Data source 2 (PPMI dataset)	The dataset was filtered to include baseline visits only (EVENT_ID = BL). Only Healthy Control and Prodromal cohorts were retained,	Cohort labels were transformed into a binary target variable (Healthy = 0, Prodromal = 1). Categorical demographic variables such as sex were encoded using binary numerical encoding. All cognitive	Cognitive-domain features representing processing speed and working memory were derived from clinical test scores. A domain-driven feature selection approach was used, retaining	Missing numerical values were handled using median imputation. Imputation statistics were calculated from the training dataset only to	Numerical features were standardized using z-score scaling. Scaling parameters were learned from the training data and applied consistently to validation and test datasets.	Explicit outlier removal was not performed. Tree-based ensemble models were used due to their robustness to moderate outliers in clinical

	while diagnosed Parkinson's disease cases were excluded. Leakage-prone variables such as cohort labels, disease stage, medication dosage (LEDD), and visit identifiers were removed.	assessment scores were ensured to be in numeric format.	only clinically relevant and mobile-feasible features such as processing speed, working memory, age, education level, and sex. Dimensionality reduction techniques such as PCA were not applied to preserve interpretability.	avoid data leakage.		cognitive data.

Objective 3

	Transformation technique						
	Data Cleaning	Data Type Conversion	Handling Subject-Specific Data	Feature Selection	Feature Engineering	Outlier Detection and Handling	Data Scaling (Standardization)
Data source 3	The dataset was examined for missing values and duplicate records. No missing entries or duplicate rows	All features were already available in appropriate numerical formats (integer and real	A subject-based train–test split was implemented to prevent data leakage inherent in longitudinal datasets. All recordings from a single subject	Feature relevance was assessed based on correlation with the total_UPDRS score and clinical importance. An optimal subset of seven features	Four composite features were derived to enhance representational capacity: voice_instability_composite, voice_quality_index,vocal_complexity_score, and age_voice_int	Explicit outlier removal or capping was not applied. Robust models such as Random Forest were selected due to their reduced sensitivity to outliers. Feature	StandardScaler was applied to all selected features, transforming them to have zero mean and unit variance. This ensured equal feature contribution

	were identified; therefore, no corrective data cleaning operations were required.	values), and no additional type conversions were necessary	were assigned exclusively to either the training set or the testing set.	(PPE, RPDE, HNR, Shimmer, Jitter (%), NHR, and age) was selected. Highly correlated and redundant features were removed to reduce multicollinearity.	eraction. These features capture combined acoustic and demographic interactions relevant to Parkinson's disease severity.	scaling was used to normalize value ranges without removing data points.	and improved performance for scale-sensitive models such as Support Vector Machines.

Objective 3

	Transformation technique						
	Data Cleaning (Audio and Text)	Data Transformation (Speech-to-Text)	Feature Engineering	Feature Selection / Dimensionality Reduction	Outlier Detection and Handling	Data Scaling (Standardization)	Data Encoding
Data source 4	Initial audio quality checks were conducted to ensure that no corrupted, silent, or critically short recordings were included. Text transcripts	Automatic Speech Recognition was performed using the openai-whisper base model to convert all audio files into text transcripts. All recordings were successfully transcribed,	A set of 29 linguistic features was extracted from each transcript, covering fluency, lexical diversity, syntactic complexity, speech patterns (e.g.,	Based on statistical significance testing (t-tests) and correlation analysis, a refined subset of 15 discriminative linguistic features was selected for modelling. This reduced dimensionality while	Outliers in selected linguistic features (e.g., words per sentence, pause frequency) were handled using the Interquartile Range (IQR) capping	StandardScaler was applied to all selected linguistic features. The scaler was fitted on the training data only and then applied to both training and test sets to prevent data leakage.	The target variable (PD/HC) was converted into numerical form using label encoding (0 and 1), enabling compatibility with machine learning

	underwent minor cleaning (e.g., removal of non-alphanumeric characters) during feature extraction.	achieving a 100% transcription success rate.	repetition ratio, filler ratio, pause frequency), and readability metrics relevant to Parkinson's disease.	preserving clinically relevant information.	method to reduce the influence of extreme values while retaining all samples.		algorithms.

Objective 4

	Transformation technique						
	DATA CLEANING	DATA TRANSFORMATION	FEATURE ENGINEERING	FEATURE SELECTION / DIMENSIONALITY REDUCTION	OUTLIER DETECTION AND HANDLING	DATA SCALING (STANDARDIZATION)	DATA ENCODING
Data source 5	Excluded trajectory files with < 2 columns or non-2D structure to remove corrupted data Auto-detected delimiters (comma, semicolon, tab, space) to handle inconsistent formatting Graceful fallback parsing used for files that failed to load, returning minimal valid arrays Class labels assigned from folder structure (healthy → 0, patient → 1) Patient IDs extracted	Image Stream Coordinate Normalization: Raw (x, y) scaled to [0, 224] canvas with 15-px padding Velocity Rendering: 224×224 grayscale images, intensity encodes velocity *(slow = bright, fast = dark), 2-px stroke width* Skeletonization: Morphological thinning to extract stroke trajectories RGB Merge: Channels combined as (Raw, Thinned, Raw) for ResNet50 compatibility Signal Stream Feature Extraction: 10D kinematic	*Image-Based Features Trajectories rendered as 224×224 velocity-encoded grayscale images Stroke intensity reflects speed (slow = bright, fast = dark) Skeletonization applied to extract stroke structure Channels merged as (Raw, Skeleton, Raw) → 3-channel RGB Pre-trained ResNet50 (frozen layers) used for feature extraction* *Output: 512D visual feature vector*	Image Based Data No explicit dimensionality reduction (e.g., PCA, t-SNE) applied Pre-trained ResNet50 performs automatic feature extraction and compression Input reduced from 224 × 224 × 3 to a compact feature representation Transfer learning with frozen early layers preserves meaningful high-level features Average + max pooling capture both global shape	Image Based Outlier Handling No explicit outlier detection applied Coordinate normalization scales all trajectories to [0, 224] canvas Standardizes drawings across scale and position differences No manual outlier removal required Velocity-encoded rendering suppresses erratic strokes *(sudden movements appear*	Image Based Normalization Pixel values rescaled from [0, 255] → [0, 1] ImageNet standardization applied: Mean = [0.485, 0.456, 0.406] Std = [0.229, 0.224, 0.225] Ensures compatibility with pre-trained ResNet50 weights Aligns input distribution with ImageNet pre-training Statistics Critical for effective transfer learning Time-Series Normalization Z-score standardization applied to all 10 kinematic features	Image Based Encoding No categorical encoding required Raw and skeletonized images converted from PIL → PyTorch tensors float32 precision used for neural network input Time-Series Label Encoding Binary label encoding applied: *Healthy Control* → 0 *Parkinson's Disease* → 1 Labels inferred from folder structure Converted to torch.LongTensor for

	from filenames (e.g., 046_HC_SPIR_R → 046_HC) Enabled spiral–wave pairing using patient IDs Included only samples with both spiral AND wave recordings for the same patient in training	features *(position, velocity, acceleration, jerk, FFT of velocity magnitude)* Normalization: Z-score standardization Windowing: 32-sample sliding windows, 75% overlap (stride = 8) Padding: Zero-padding applied to incomplete final windows Window Selection: Middle window used as final signal representation Tensor Format: Transposed from (Time, Features) → (Features, Time) for 1D CNN input	Time-Series Features 10D kinematic features extracted from *(x, y)* at 60 Hz Features include position, velocity, acceleration, jerk FFT of velocity magnitude captures 4–6 Hz tremor Features processed via CNN–BiGRU Output: 128D temporal feature vector	and local stroke details Time-Series Data All engineered kinematic features retained due to clinical relevance to PD Features include position, velocity, acceleration, jerk, and frequency-domain metrics No manual feature selection applied CNN-BiGRU automatically learns salient temporal patterns	*darker and less visually dominant)* TimeSeries Outlier Handling No explicit outlier detection Trajectories with < 2 data points excluded during data cleaning Z-score normalization reduces influence of extreme values Future enhancement: Velocity-based filtering *(remove points > 3σ from mean velocity to handle noise/glitches)*		CrossEntropy Loss Label smoothing (0.1) applied to reduce overconfidence Improves generalization and training stability

3.2. Scalability

Objective 1

The PADS dataset demonstrates exceptional sufficiency and scalability for the development of the Edge-AI tremor acquisition system. It provides significant statistical power with data from 469 participants, including a large cohort of Parkinson's Disease patients, which far exceeds the sample size of typical wearable pilot studies. This volume mitigates the risk of overfitting and ensures the model is trained on a diverse range of motor patterns, including differential diagnoses.

In terms of sufficiency, the dataset's high temporal resolution of 100 Hz captures the granular kinematic details necessary to isolate subtle pathological tremors (typically 4–6 Hz) from voluntary movements. Its scalability is evident in the standardized data structure, which allows for efficient down-sampling and feature extraction—critical for optimizing TinyML inference on resource-constrained devices like the ESP32. Furthermore, the inclusion of clinical MDS-UPDRS Part III scores enables the development of regression-based models for severity estimation, providing a robust framework that supports both binary detections today and more complex disease progression monitoring in the future.

Objective 2

The scalability of the selected dataset was evaluated in terms of data volume, feature dimensionality, and suitability for both offline analysis and real-world deployment.

The PPMI cognitive dataset contains longitudinal cognitive assessments collected over multiple years, with many participants across different clinical cohorts. For this research component, only baseline cognitive assessments were used, resulting in a sufficiently large and balanced dataset for supervised learning and statistical analysis.

The dataset size is scalable for machine learning experimentation, as it supports:

- Robust feature selection and model training without overfitting
- Cross-validation and repeated experimentation
- Extension to longitudinal analysis in future work

Additionally, the selected cognitive features are low-dimensional and computationally efficient, making them suitable for integration into mobile and cloud-based systems. This ensures that the proposed approach can scale from offline research experiments to real-time cognitive screening applications.

Objective 3

The acoustic and linguistic datasets demonstrate complementary strengths for the AI-powered speech analysis system. The Oxford Parkinson's Disease Telemonitoring Dataset provides exceptional sufficiency with 5,875 recordings from 42 subjects, averaging 139.9 recordings per individual. Its longitudinal structure enables robust tracking of disease progression and within-subject variability, while acoustic feature extraction remains computationally efficient and highly scalable. The MDVR KCL Dataset, though smaller with 73 recordings, offers valuable task diversity through ReadText and SpontaneousDialogue components, facilitating comprehensive linguistic biomarker identification. Its scalability lies in the efficient ASR-based transcription pipeline using Whisper and extraction of 15 well-defined linguistic features, ensuring interpretable and clinically relevant models. The acoustic dataset excels in volume and regression modeling capability, while the linguistic dataset provides essential task-based diversity and explainability. Together, these datasets establish a robust, scalable framework where acoustic depth complements linguistic interpretability, supporting both current analysis needs and future system expansion

Objective 4

The Smartphone Finger-Drawing Touch-Motion Dataset contains 58 subjects (30 healthy controls and 28 Parkinson's patients), providing 116 paired samples when considering both spiral and wave drawings per subject. While this dataset size is sufficient for initial model validation and proof-of-concept development, it presents limitations for production-scale deployment and generalization across diverse populations. To address scalability constraints, 10-fold stratified cross-validation was employed to maximize training data utilization while ensuring robust performance estimation across all available samples. Extensive data augmentation techniques were implemented including image-based augmentations (AugMix with rotation, shear, translation, brightness, and contrast adjustments) and signal-based augmentations (time stretching by 0.9-1.1x factor and Gaussian noise injection with σ=0.01), effectively increasing the training data diversity without requiring additional data collection. The sliding window approach with 75% overlap further multiplies the number of training samples from the time-series data. Transfer learning with a pre-trained ResNet50 backbone reduces the data requirements by leveraging features learned from millions of ImageNet images. The model architecture with 26.8 million total parameters (18.3 million trainable) is computationally efficient, requiring approximately 15-20 minutes per fold on an A100 GPU, making it scalable for larger datasets. For future scalability, the pipeline is designed to seamlessly incorporate additional datasets such as NewHandPD (92 subjects) for cross-dataset validation, enabling assessment of generalization performance across different populations and data collection protocols.

3.3. Feature extraction

Objective 1

Feature extraction was carried out to identify motor biomarkers that are both clinically discriminative and suitable for real-time wearable-based Parkinson's disease tremor detection. First, a comprehensive pool of kinematic features was formed using raw accelerometer and gyroscope sensor data from the PADS (Parkinson's Disease Smartwatch Dataset). These features covered multiple signal domains including time-domain statistics, frequency-domain spectral characteristics, and tremor-specific power metrics derived from both wrist sensors across 11 standardized movement tasks.

To ensure meaningful and non-redundant feature representation, raw 3-axis sensor readings were transformed into rotation-invariant magnitude signals using vector summation. Bandpass filtering (0.5–20 Hz) was applied to remove gravitational offset and high-frequency noise artifacts. This preprocessing reduced sensor orientation dependency and improved signal quality for downstream analysis.

Signal processing–based feature engineering was then performed using sliding window segmentation (2.56 seconds, 50% overlap) to extract 16 discriminative features per window. The most influential features consistently belonged to:

- **Tremor-band power (3–7 Hz)**: Captures the characteristic Parkinsonian rest tremor frequency range
- **Spectral entropy**: Quantifies signal regularity, as tremor exhibits more rhythmic patterns than healthy movement
- **RMS and standard deviation**: Measures movement intensity and variability
- **Dominant frequency**: Identifies primary oscillation characteristics

Based on this analysis, a refined feature set of 16 attributes (8 accelerometer + 8 gyroscope) was selected that balances predictive power and computational feasibility for TinyML deployment on resource-constrained microcontrollers. These features were further optimized for real-time extraction on an ESP32 device, enabling equivalent on-device computation of tremor probability using a quantized neural network model.

This structured feature extraction strategy ensured that the final model inputs were both clinically meaningful and hardware-optimized, supporting integration into a wearable-based early Parkinson's tremor screening system with real-time inference capability.

Objective 2

Feature extraction was carried out to identify cognitive attributes that are both clinically discriminative and suitable for mobile-based cognitive assessment.

First, a comprehensive pool of cognitive features was formed using standardized neuropsychological test scores available in the PPMI dataset. These features covered multiple cognitive domains including global cognition, processing speed, executive function, memory, attention, fluency, and visuospatial ability.

To ensure meaningful and non-redundant feature representation, variants of the same cognitive tests (e.g., raw scores, derived scores, and standardized versions) were analyzed, and only the primary raw clinical scores were retained. This reduced multicollinearity and improved interpretability.

Machine learning–based feature importance analysis was then performed using ensemble models to quantify the contribution of each cognitive feature toward Parkinson's disease discrimination. The most influential features consistently belonged to:

- **Processing speed** (e.g., symbol–digit based measures)
- **Executive function** (e.g., trail-making and sequence-based tasks)
- **Working memory and attention**
- **Semantic and phonemic fluency**

Based on this analysis, a refined feature set was selected that balances predictive power and feasibility for mobile implementation. These features were further mapped to simplified digital cognitive tasks, enabling future extraction of equivalent mobile-derived features such as reaction time, error rates, and response variability.

This structured feature extraction strategy ensured that the final model inputs were both data-driven and application-oriented, supporting integration into a mobile-based early cognitive screening system.

Objective 3

Feature extraction was performed to identify and evaluate the key acoustic and linguistic attributes relevant to early detection of Parkinson's Disease, in alignment with the stated research objective. Separate feature extraction strategies were applied for acoustic and linguistic datasets.

Acoustic Feature Extraction

For the acoustic modality, the dataset provided pre-extracted biomedical voice features, therefore feature extraction focused on evaluating and selecting the most relevant attributes rather than deriving new features.

Initially, the dataset contained multiple acoustic measures related to jitter, shimmer, pitch variability, noise characteristics, and non-linear dynamics, along with demographic information such as age. Feature relevance was evaluated using correlation analysis with the total_UPDRS score, which represents Parkinson's disease severity.

To address multicollinearity, highly correlated acoustic features were examined, and representative features were retained from each correlated group. Feature selection also considered clinical relevance, prioritizing well-established Parkinson's speech biomarkers.

Based on statistical relevance and clinical importance, a final set of seven acoustic features was selected: PPE, RPDE, HNR, Shimmer, Jitter (%), NHR, and age. These features were used as inputs for the acoustic modelling stage.

Linguistic Feature Extraction

For the linguistic modality, feature extraction involved a multi-stage process starting from raw audio recordings.

First, all audio samples were converted into text transcripts using an Automatic Speech Recognition (ASR) system. This enabled Natural Language Processing (NLP) techniques to be applied for linguistic analysis.

From each transcript, a comprehensive set of linguistic features was extracted to capture speech characteristics associated with Parkinson's disease. These features included:

- Fluency and speech rate measures (e.g., words per sentence, pause frequency)
- Lexical diversity metrics (e.g., vocabulary richness, word length)
- Syntactic and grammatical patterns (e.g., part-of-speech ratios)
- Speech pattern indicators such as repetition and fillers
- Readability scores reflecting language complexity

In total, 29 linguistic features were initially extracted.

Feature evaluation was conducted using statistical significance testing (t-tests) and correlation analysis with Parkinson's disease status (PD vs. HC). Based on these analyses, a refined set of 15 discriminative linguistic features was selected. This reduced dimensionality while retaining the most informative linguistic markers for classification.

Objective 4

Image-Based Feature Extraction: For visual feature extraction, a pre-trained ResNet50 model with ImageNet weights was employed as the backbone. Early convolutional layers (conv1, bn1, layer1, layer2, layer3) were frozen to preserve low-level and mid-level visual features learned from large-scale image classification, while layer4 remained trainable to adapt high-level features for drawing analysis. A dual pooling mechanism combining Global Average Pooling and Global Max Pooling was applied to the final convolutional output, capturing both global context and salient local patterns. The concatenated pooling outputs were projected through a fully connected layer to produce a 512-dimensional feature vector representing the visual characteristics of each drawing, including stroke patterns, line consistency, tremor-induced wobbles, and overall drawing quality.

Time-Series Feature Extraction: For temporal feature extraction, a Multi-Scale 1D CNN-BiGRU architecture was implemented. The CNN component consists of two multi-scale convolutional blocks, each with parallel branches using kernel sizes of 1, 3, and 5 to capture patterns at different temporal resolutions, along with a max-pooling branch for local peak detection. Residual connections were added to prevent gradient degradation. The outputs of all branches were concatenated and passed through batch normalization and ReLU activation. The processed features were then fed into a Bidirectional GRU layer with 64 hidden units per direction, enabling the model to capture both forward and backward temporal dependencies in the kinematic sequences. A 4-head self-attention mechanism was applied to the GRU outputs to emphasize the most informative time steps. The final hidden states from both GRU directions were concatenated and projected through a dropout layer and fully connected layer to produce a 128-dimensional temporal feature vector capturing
movement dynamics, velocity patterns, and tremor signatures.

4. Modelling and Results

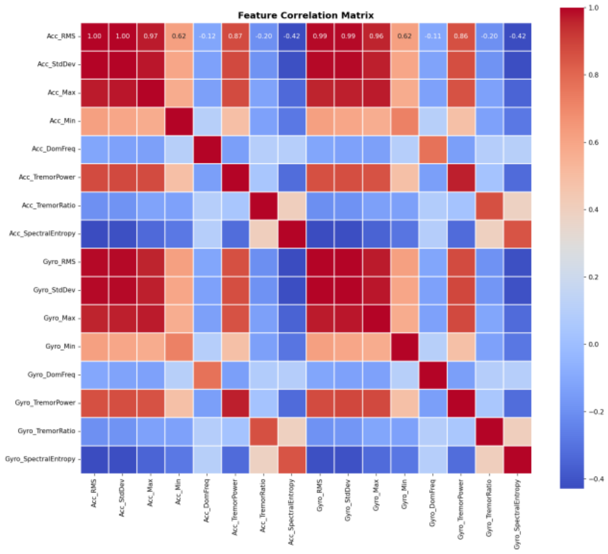
4.1. Key Insights:

<u>**PADS Parkinson's Disease Smartwatch Dataset**</u>
Analysis of the wearable sensor dataset demonstrated that accelerometer and gyroscope-derived kinematic features are strong indicators of Parkinson's disease tremor presence. Exploratory analysis revealed that the dataset contained recordings from multiple subjects across 11 standardized movement tasks, with sensor data captured from both left and right wrist smartwatches at a sampling rate of 100 Hz.

1. Feature Correlation Analysis

Correlation heatmaps and feature relationship plots showed that tremor-band power features (3–7 Hz) and spectral characteristics, including Acc_TremorPower, Gyro_TremorPower, Acc_SpectralEntropy, and Gyro_SpectralEntropy, were strongly associated with Parkinson's disease classification. RMS and standard deviation features from both sensors also showed moderate discriminative power, highlighting the importance of movement intensity and variability in tremor detection.
The correlation matrix below reveals strong positive correlations between accelerometer features (Acc_RMS, Acc_StdDev, Acc_Max all showing $r > 0.97$) and

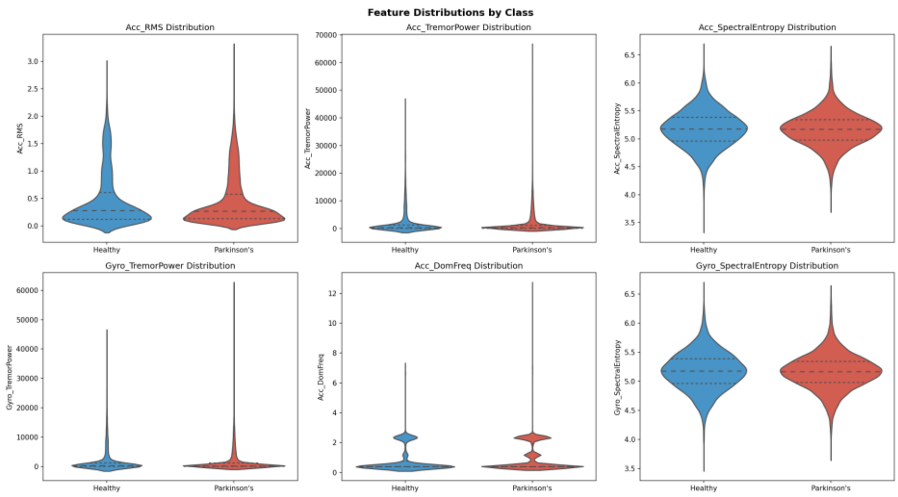
similarly between gyroscope features (Gyro_RMS, Gyro_StdDev, Gyro_Max with r > 0.96). Tremor power features show moderate correlation with intensity measures, while spectral entropy features demonstrate negative correlation with tremor power, consistent with the rhythmic nature of Parkinsonian tremor.



2. Feature Distribution Comparison

Violin plot distributions comparing Healthy versus Parkinson's classes revealed clear separation in tremor-specific features, with Parkinson's subjects exhibiting higher tremor-band power and lower spectral entropy values, consistent with the rhythmic nature of Parkinsonian rest tremor.
Key observations from the feature distributions:

- Acc_RMS: Healthy subjects show tighter distribution around lower values, while Parkinson's subjects exhibit higher intensity with wider variance
- Acc_TremorPower & Gyro_TremorPower: Parkinson's subjects show notably higher power in the 3-7 Hz tremor band, creating clear separation between classes
- Spectral Entropy: Healthy subjects demonstrate higher entropy (more random movement patterns), while Parkinson's subjects show lower entropy (more rhythmic/periodic tremor patterns)
- Acc_DomFreq: Both classes show similar dominant frequency distributions, suggesting frequency alone is not sufficient for classification

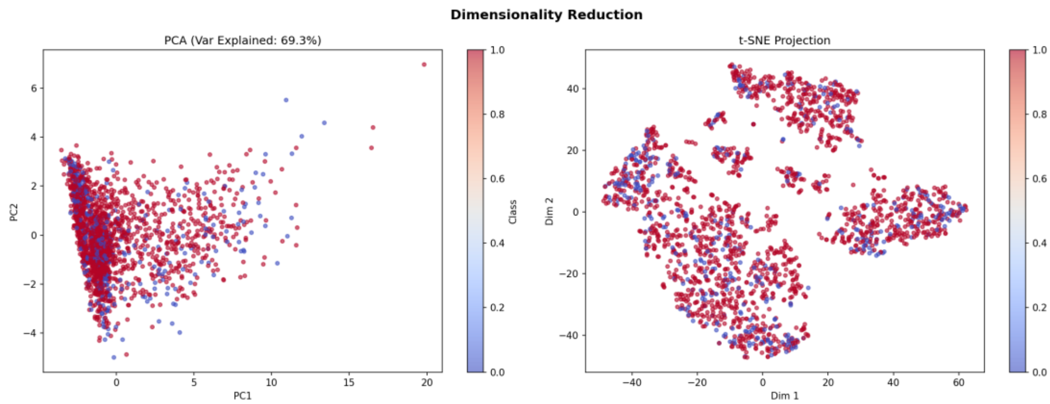


3. Dimensionality Reduction Visualization

Dimensionality reduction using PCA and t-SNE projections confirmed visible clustering between the two classes, validating the discriminative capacity of the engineered feature set.

PCA Analysis: Principal Component Analysis with 2 components explained 69.3% of the total variance in the feature space. The scatter plot shows partial linear separability between Healthy (blue) and Parkinson's (red) classes, with some overlap in the central region indicating the complexity of the classification task.

t-SNE Analysis: The t-SNE projection reveals more distinct cluster formations, with Parkinson's samples (red) forming tighter groupings compared to the more dispersed Healthy samples (blue). This non-linear visualization demonstrates that the 16-feature representation captures meaningful tremor patterns that separate the two classes in the underlying data manifold.

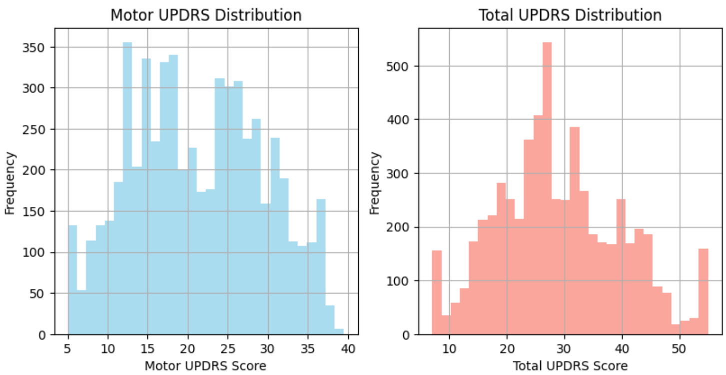


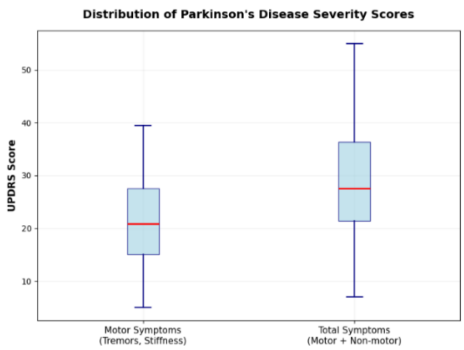
Summary

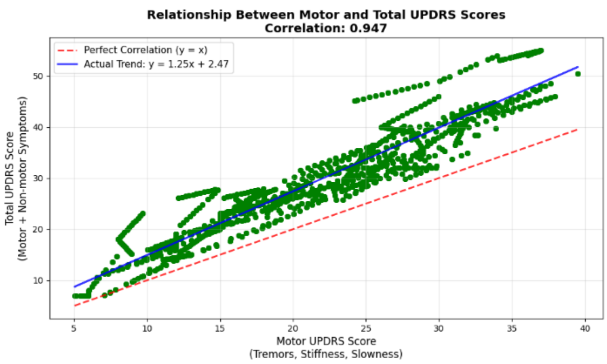
Feature importance analysis through model training and class distribution visualization revealed that tremor-band power features were the most influential, followed by spectral entropy and RMS-based intensity measures. This confirms that both frequency-domain tremor signatures and time-domain movement characteristics contribute meaningfully to Parkinson's tremor detection. The final 16-feature set was validated for TinyML deployment, ensuring real-time inference capability on resource-constrained ESP32 microcontrollers.

<u>Oxford Parkinson’s Disease Telemonitoring Dataset</u>

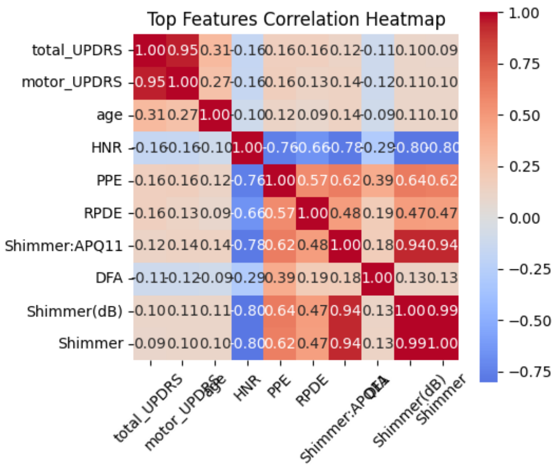
Analysis of the acoustic dataset demonstrated that speech-based acoustic features are strong indicators of Parkinson’s disease severity. Exploratory analysis revealed a near-perfect correlation between motor_UPDRS and total_UPDRS, confirming that modelling vocal markers related to motor symptoms can effectively represent overall disease progression.







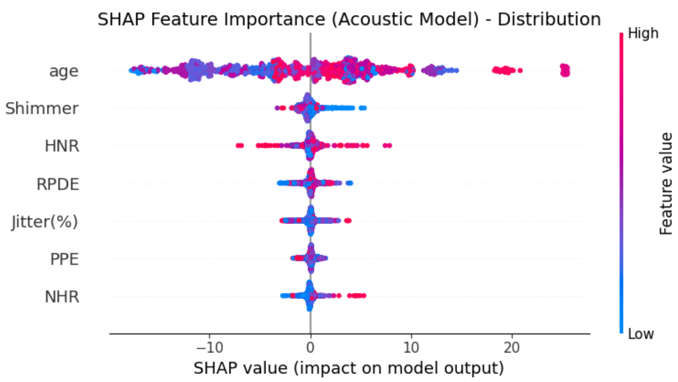
Correlation heatmaps and feature relationship plots showed that voice instability and noise-related features, including PPE, RPDE, Jitter (%), Shimmer, NHR, and HNR, were strongly associated with disease severity. Age also showed a moderate positive relationship with UPDRS scores, highlighting its influence on progression.



A Random Forest Regressor trained on the selected acoustic features achieved strong predictive performance, explaining a large proportion of variance in disease severity (R^2 = 0.825, MAE = 3.086, RMSE = 4.400).
These results indicate that acoustic speech features can reliably estimate Parkinson's disease severity in a continuous manner.

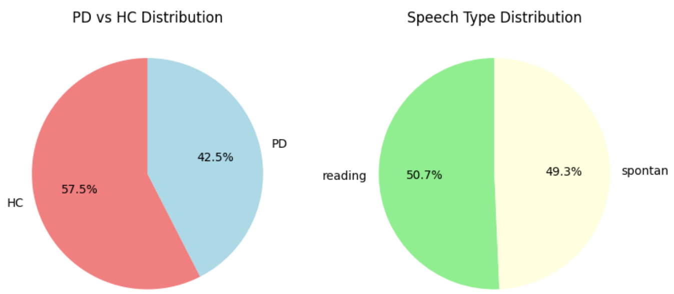
```
RANDOM FOREST RESULTS:
  Mean Absolute Error (MAE):   3.086
  Root Mean Squared Error (RMSE): 4.400
  R² Score:                    0.825
```

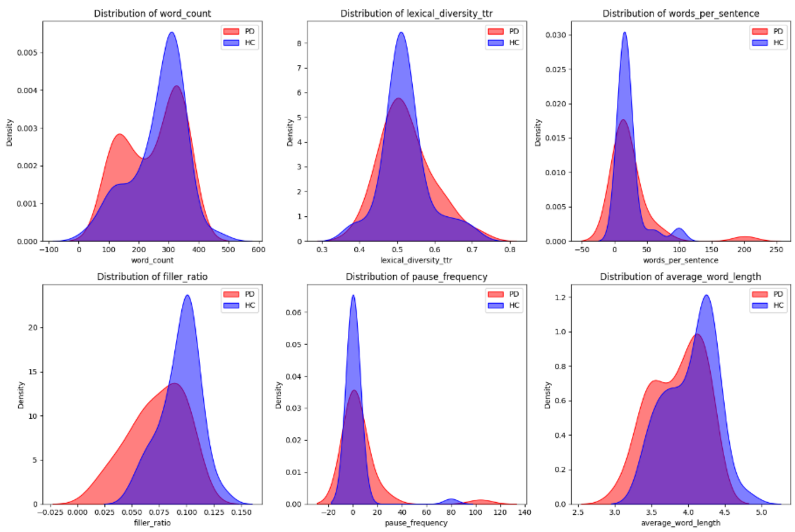
Feature importance analysis using SHAP values revealed that age was the most influential feature, followed by acoustic instability measures such as Shimmer and Jitter (%), and voice quality indicators such as HNR and RPDE. This confirms that both demographic and physiological speech markers contribute meaningfully to severity prediction.



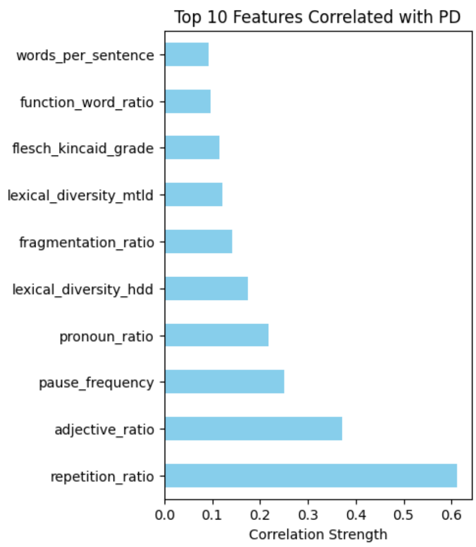
MDVR KCL Dataset

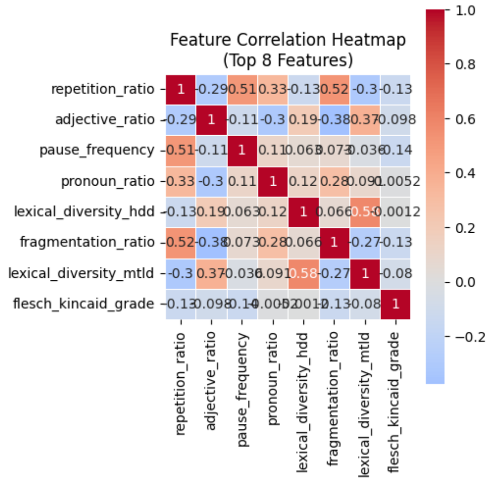
Exploratory analysis of the linguistic dataset revealed clear differences between Parkinson's disease and healthy control speech. Visualizations showed that Parkinson's disease speech was characterized by higher repetition rates, increased pause frequency, and reduced lexical complexity, particularly in spontaneous speech.



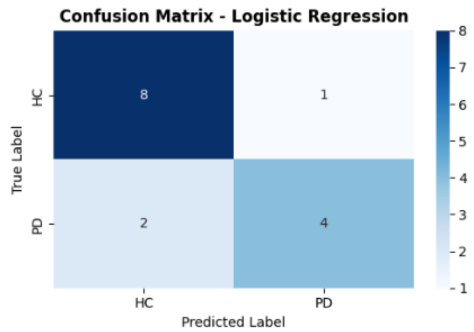


Statistical testing confirmed several linguistically significant features differentiating PD and HC groups. Correlation analysis further highlighted repetition_ratio as the most strongly associated linguistic marker.



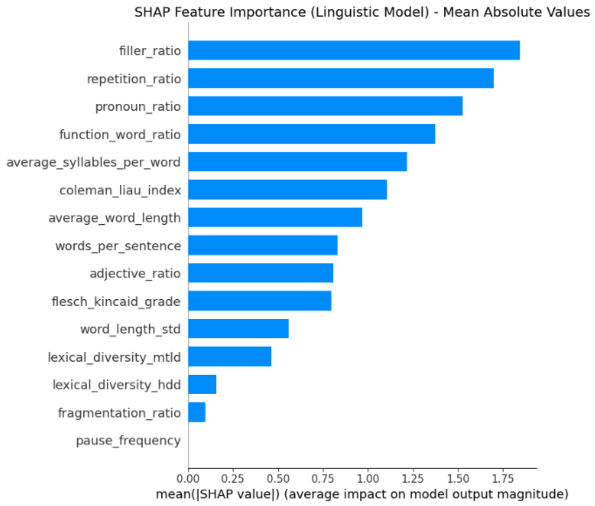


A Logistic Regression classifier trained on selected linguistic features demonstrated promising classification performance, achieving a test accuracy of 0.80 and F1-score of 0.796 before tuning. Cross-validation results indicated stable generalization performance (mean CV accuracy = 0.758), supporting the reliability of the linguistic model despite the small dataset size.




```
BEST PERFORMING MODEL: Logistic Regression
  Accuracy: 0.8000
  F1-Score: 0.7962
```

SHAP-based feature interpretation showed that repetition_ratio had the strongest positive influence on Parkinson’s disease prediction, followed by features related to function word usage and readability. Some features, such as filler_ratio, showed unexpected negative influence, highlighting areas for further linguistic investigation and dataset expansion.

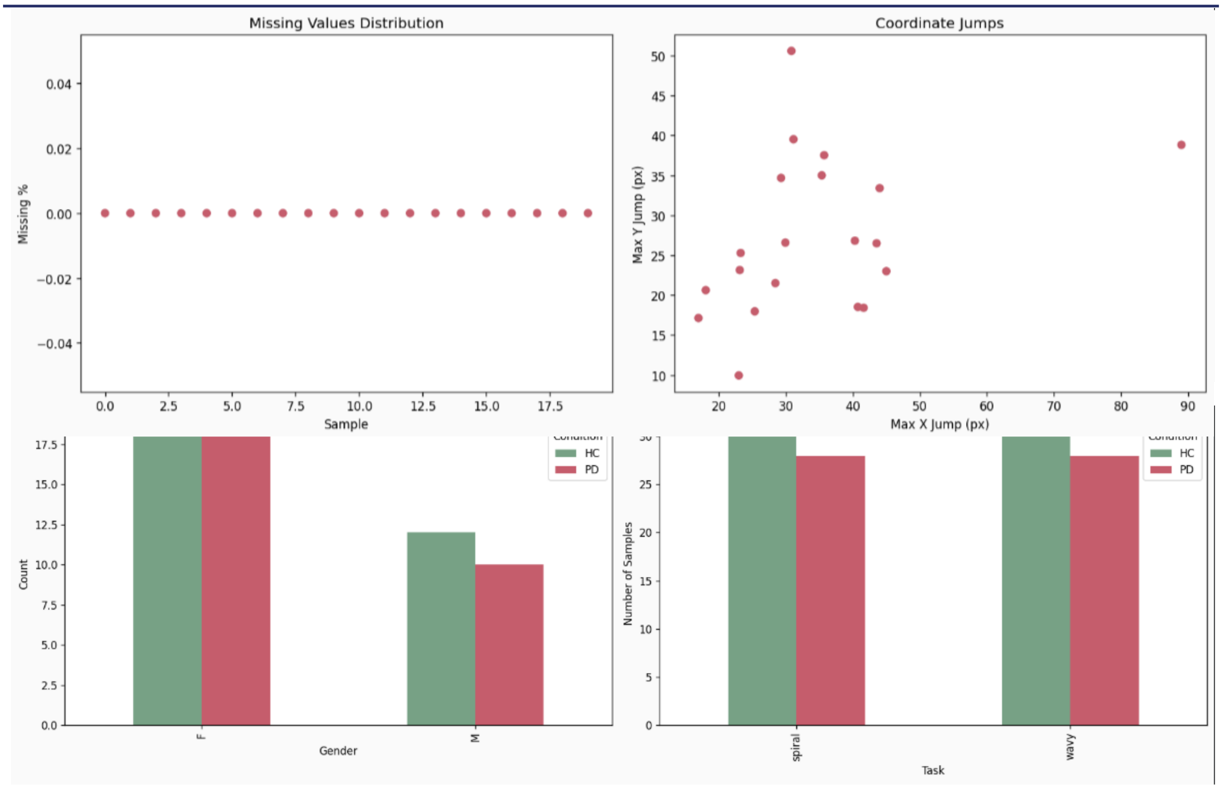


Smartphone Finger-Drawing Parkinson’s Touch-Motion Dataset

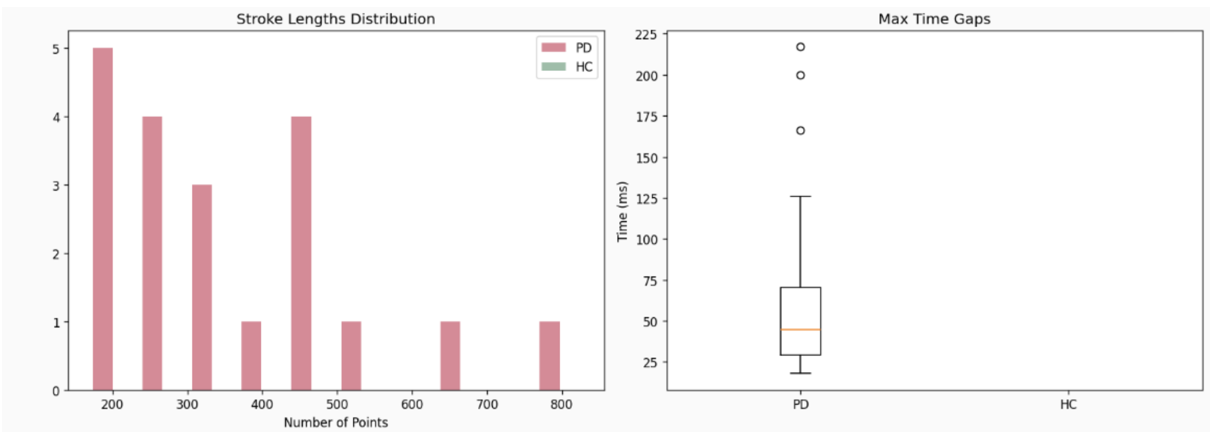
The analysis of the Smartphone Finger-Drawing Parkinson’s Touch-Motion Dataset reveals clear and consistent differences between Parkinson’s Disease (PD) and Healthy Control (HC) subjects across demographic, temporal, spatial, kinematic, and frequency-domain characteristics. These findings validate the effectiveness of smartphone-based drawing tasks as a source of digital motor biomarkers for early Parkinson’s disease detection.

Demographic and Data Quality Insights

Exploratory analysis confirmed a near-balanced subject distribution between PD and HC groups, with comparable age ranges and gender composition. This demographic consistency reduces confounding effects and ensures that observed differences in movement patterns are primarily attributable to disease-related motor impairments rather than population bias. Data quality analysis further showed negligible missing values, consistent stroke lengths, and well-formed trajectories across samples, confirming the suitability of the dataset for deep learning–based modelling without extensive imputation.



These plots collectively highlight data quality and motion irregularities, with PD samples exhibiting greater temporal and spatial variability than HC samples.

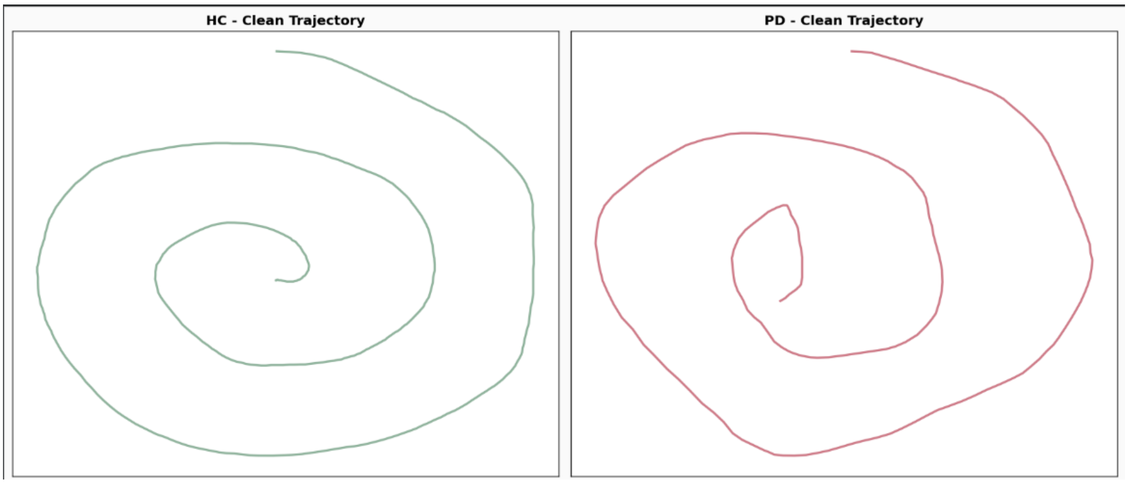


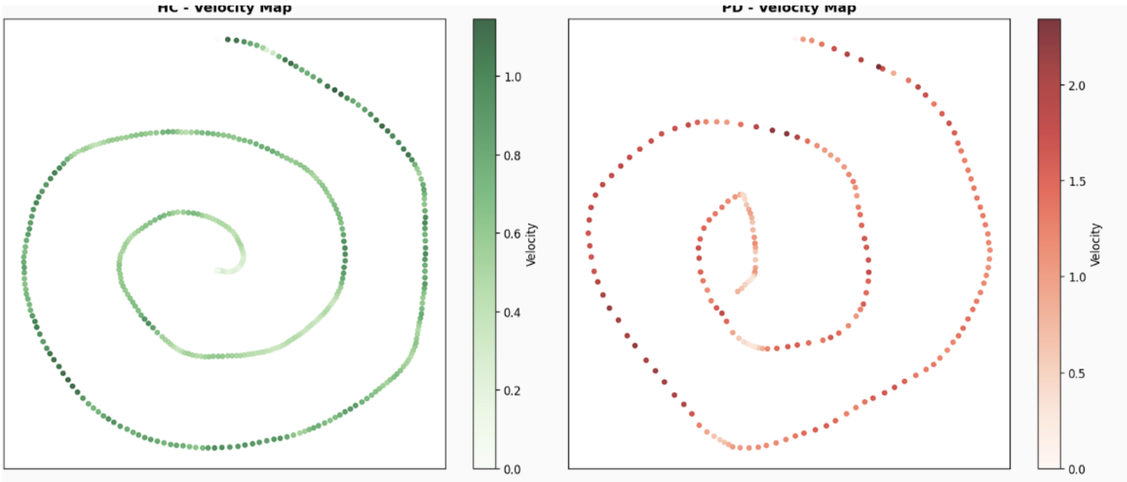
Temporal Motion Characteristics

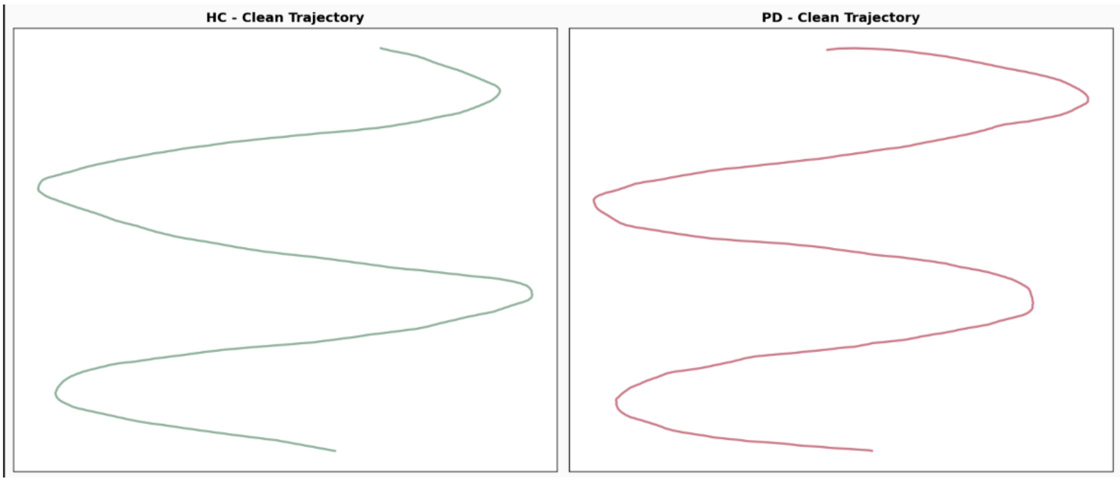
Time-series visualizations of X- and Y-coordinates, velocity, and acceleration demonstrated distinct motion behaviors between PD and HC subjects. HC subjects exhibited smooth, continuous trajectories with stable velocity and low acceleration variability. In contrast, PD subjects showed irregular positional fluctuations, frequent velocity spikes, and pronounced acceleration bursts, indicating reduced motor stability and impaired movement smoothness. These temporal irregularities reflect clinically recognized Parkinsonian symptoms such as bradykinesia and tremor-induced hesitation.

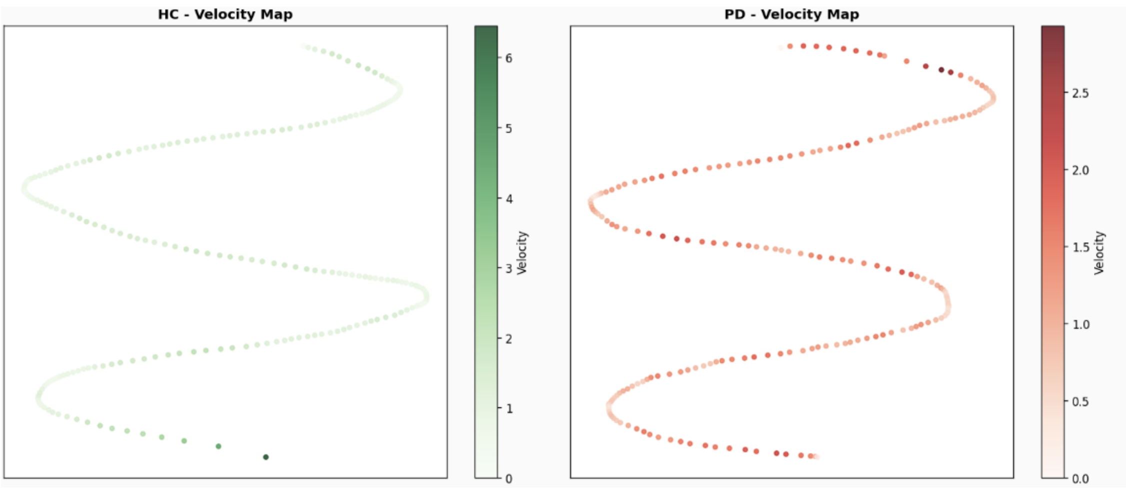
Spatial Trajectory and Velocity Patterns

Clean trajectory plots revealed that HC drawings closely followed smooth and well-proportioned spiral and wave patterns. PD trajectories, however, appeared distorted, uneven, and less symmetric, with visible deviations from the intended shapes. Velocity-encoded trajectory maps further highlighted these differences, where HC samples showed gradual and consistent velocity distributions, while PD samples exhibited localized high-velocity regions, suggesting tremor-like oscillations and abrupt corrective movements.



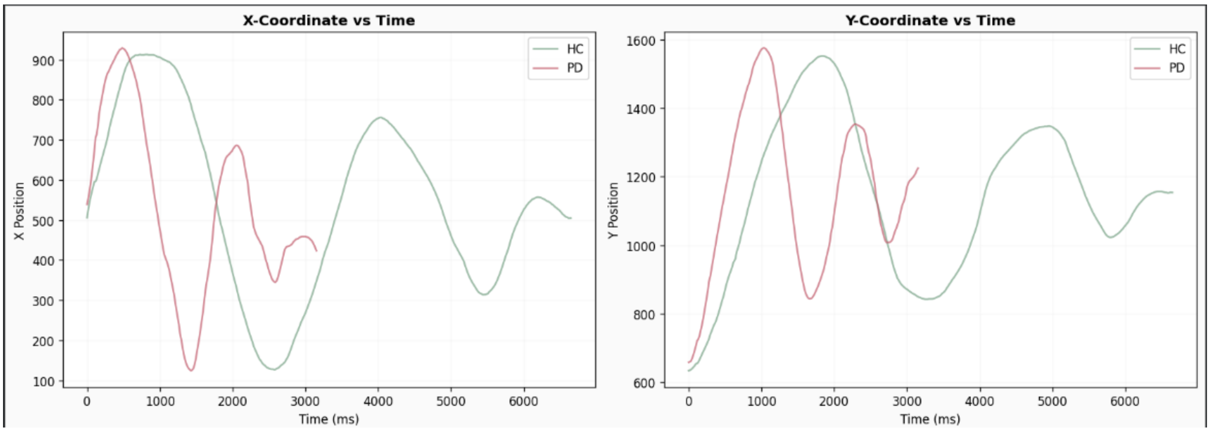

HC - Velocity Map
PD - Velocity Map
Velocity
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0

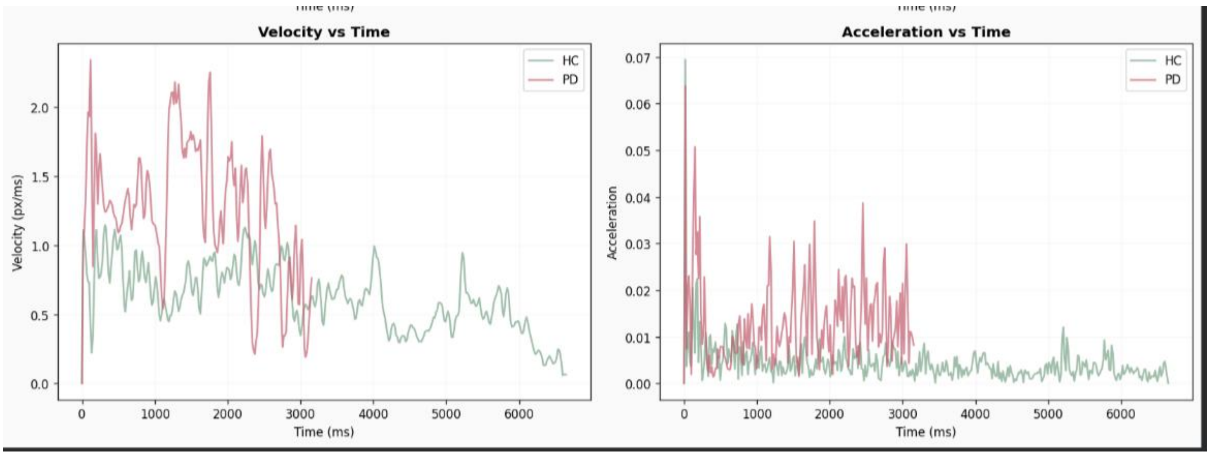

HC - Clean Trajectory
PD - Clean Trajectory

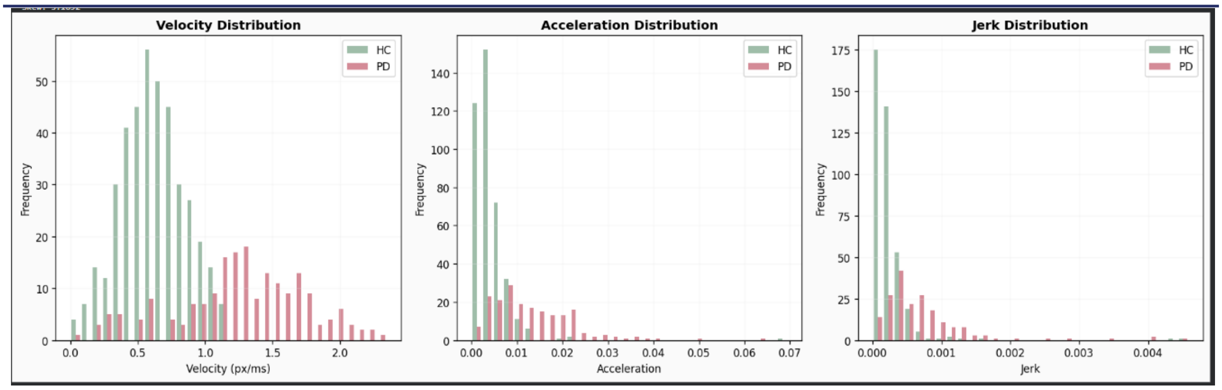

HC - Velocity Map
PD - Velocity Map
Velocity
0
1
2
3
4
5
6
0.0
0.5
1.0
1.5
2.0
2.5

Kinematic Feature Distributions

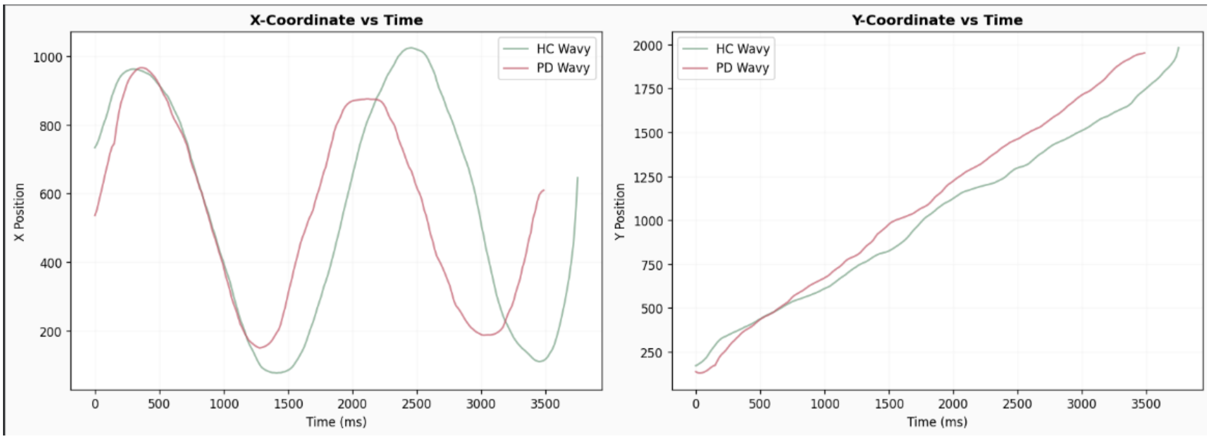
Statistical analysis of kinematic features showed that PD subjects had higher mean and variance in velocity, acceleration, and jerk compared to HC subjects. Distribution plots revealed strong right-skewness in PD acceleration and jerk, indicating frequent sudden changes in motion.
These higher-order kinematic features effectively capture movement smoothness and motor instability, making them highly discriminative for Parkinson's disease classification.



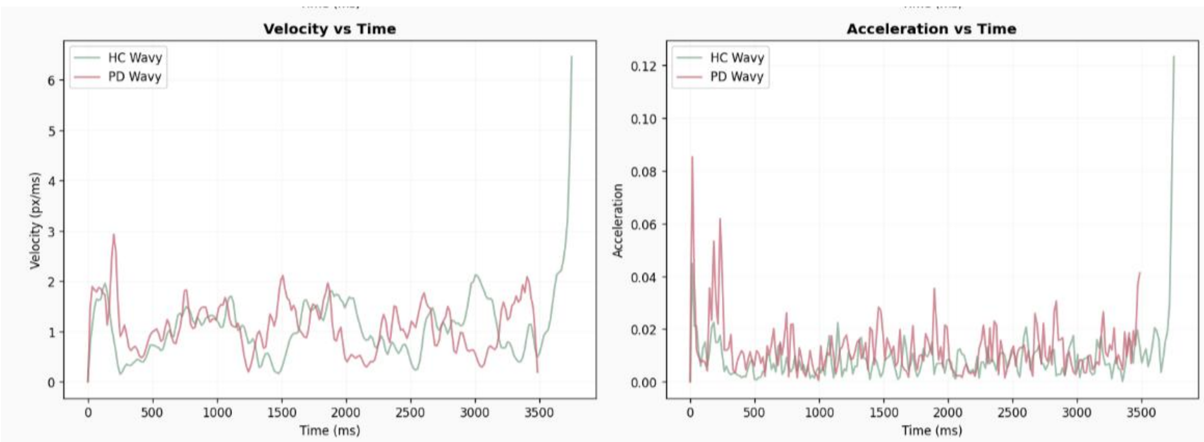




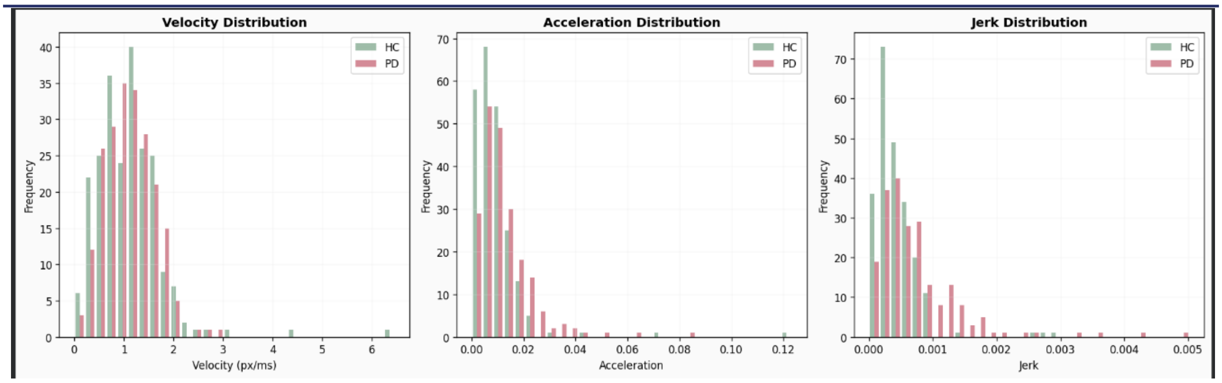
Velocity Distribution
HC
PD
Frequency
Velocity (px/ms)
Acceleration Distribution
Acceleration
Jerk Distribution
Jerk



X-Coordinate vs Time
HC Wavy
PD Wavy
X Position
Time (ms)
Y-Coordinate vs Time
Y Position

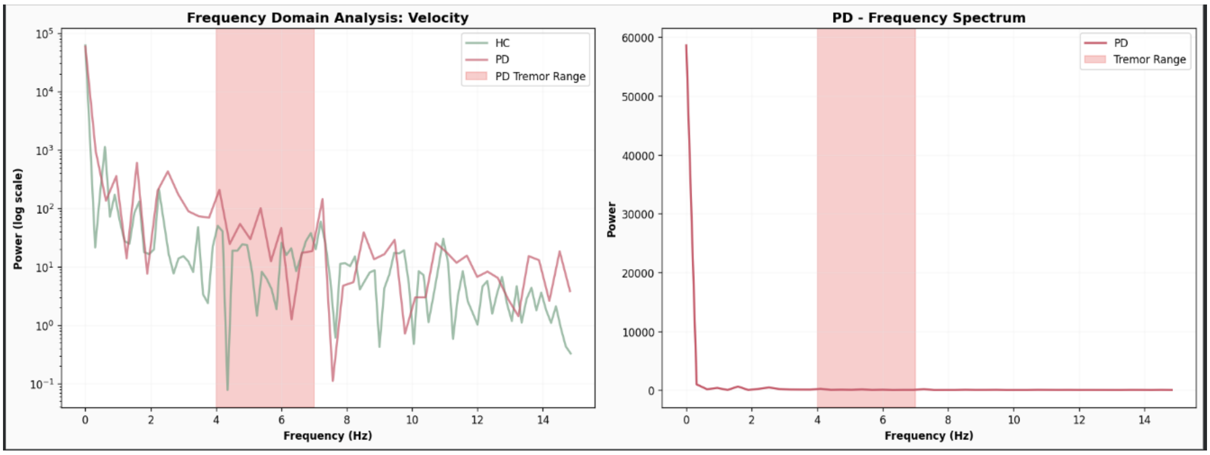


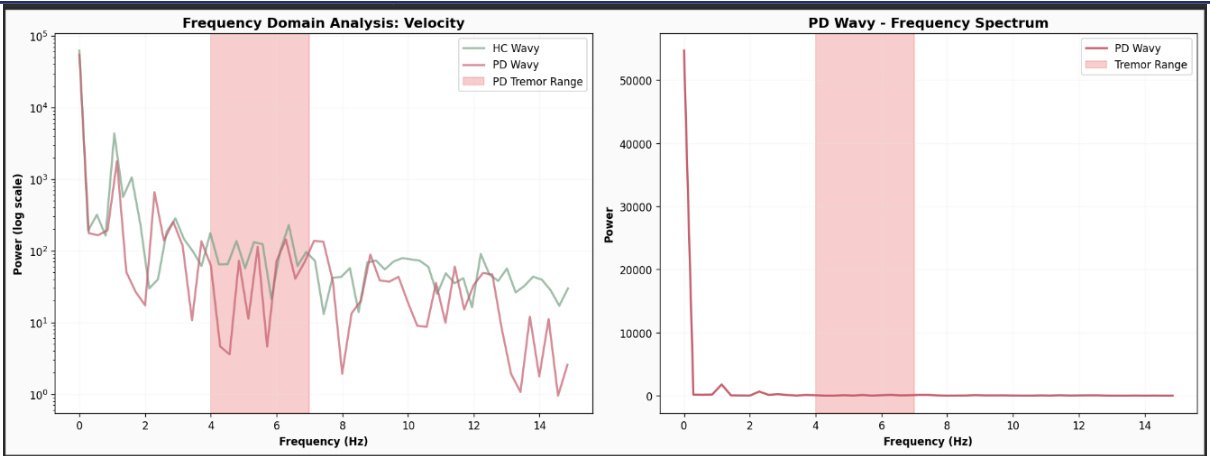
Velocity vs Time
HC Wavy
PD Wavy
Velocity (px/ms)
Time (ms)
Acceleration vs Time
Acceleration



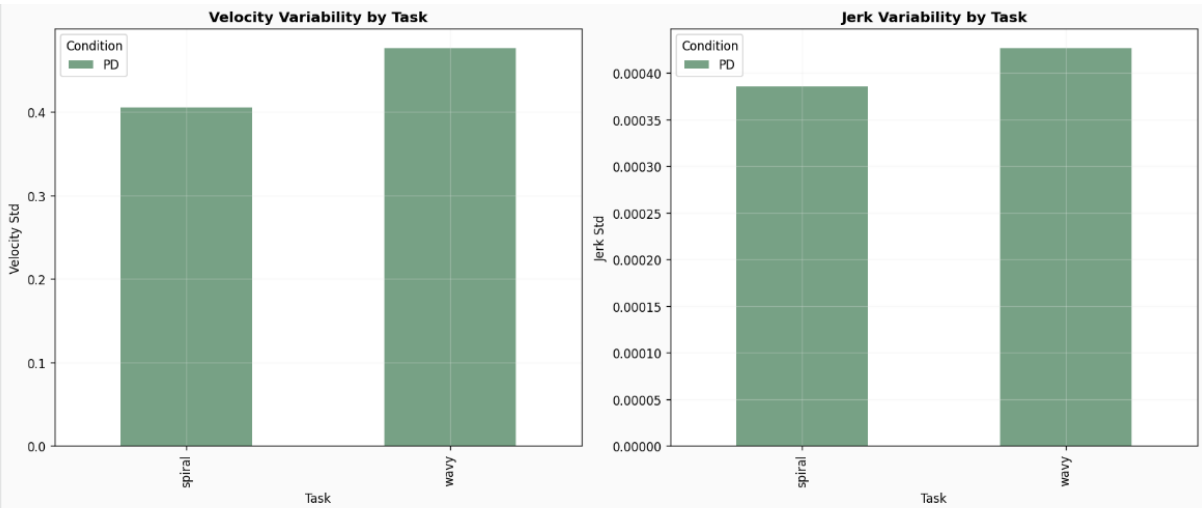
Frequency-Domain Insights

Frequency-domain analysis of velocity signals using Fast Fourier Transform (FFT) revealed elevated spectral power in the 4–7 Hz range for PD subjects, corresponding to the clinically established Parkinsonian tremor band. HC samples showed significantly lower power within this frequency range, reflecting stable and tremor-free motion.
The presence of dominant tremor-band peaks in PD samples confirms that frequency-domain features provide complementary and clinically meaningful information beyond time-domain kinematics.





Task-level comparison between spiral and wave drawings showed that the wavy task consistently produced higher velocity and jerk variability, particularly in PD subjects. This suggests that increased task complexity amplifies motor instability, making the wave task more sensitive to subtle Parkinsonian impairments.
These findings justify the inclusion of both tasks in the Quad-Stream architecture, as they capture complementary motor characteristics.



Overall, the exploratory and statistical analyses demonstrate that PD subjects consistently exhibit increased temporal irregularity, spatial distortion, kinematic variability, and tremor-related frequency components compared to healthy controls. These patterns are consistently observed across both spiral and wave tasks. The extracted visual, kinematic, and frequency-domain features therefore provide a strong and clinically grounded foundation for the proposed Quad-Stream multimodal deep learning framework for early Parkinson’s disease detection.

Quad-Stream Model Performance and Analysis

A Multi-Modal Quad-Stream Architecture (CNN-BiGRU + ResNet50) was trained using fused image-based drawing representations and time-series kinematic features extracted from smartphone finger-drawing tasks. The model integrates spatial visual patterns from spiral and wave drawings with temporal motion dynamics, enabling comprehensive characterization of Parkinson's disease–related motor impairments.

Training and Validation Behavior

The training and validation loss curves show a consistent downward trend, indicating stable convergence without severe overfitting. Correspondingly, the training and validation accuracy curves demonstrate gradual improvement across epochs, confirming effective learning from the fused multimodal inputs. The One-Cycle learning rates scheduler further contributed to faster convergence and stable optimization.

Classification Performance

Confusion matrix analysis indicates strong discrimination between Parkinson's Disease (PD) and Healthy Control (HC) subjects, with a high proportion of correctly classified samples across both classes. Class-wise performance metrics reveal balanced precision, recall, and F1-scores, suggesting that the model does not disproportionately favor either class despite the limited dataset size.

Cross-Validation Results

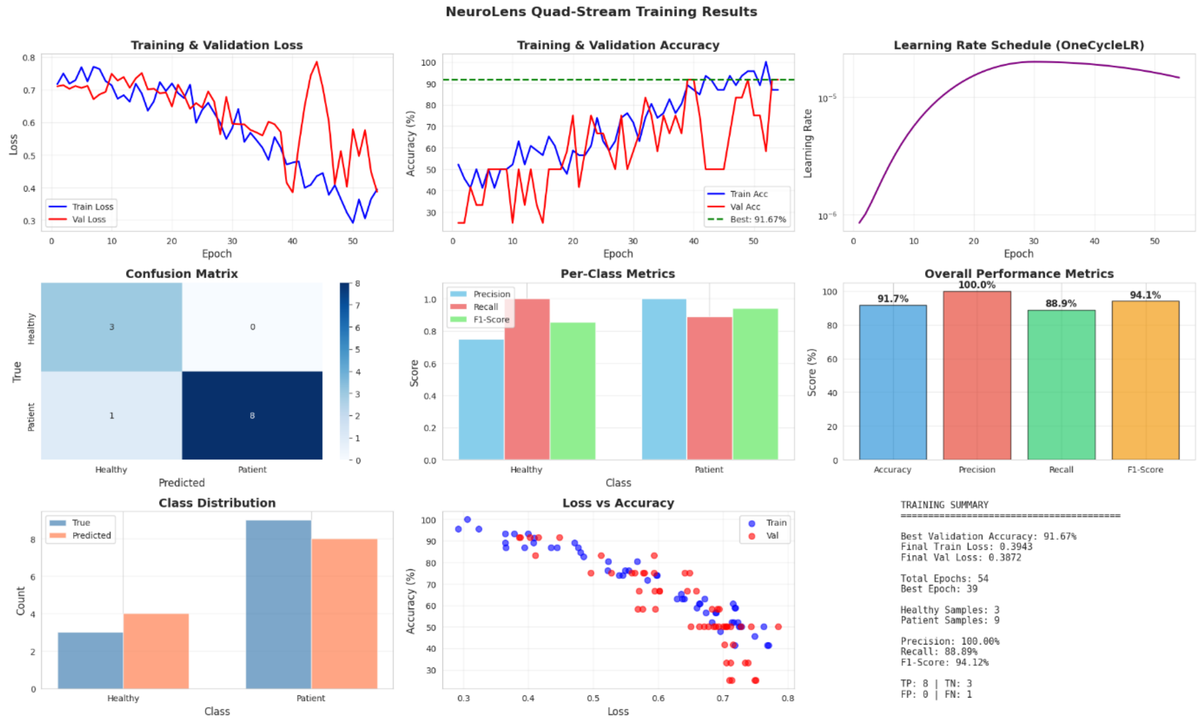
Ten-fold cross-validation analysis highlights variability across folds, which is expected given the small sample size. However, the mean accuracy and F1-score remain consistently high, demonstrating the robustness of the Quad-Stream architecture. Sensitivity and specificity analysis across folds further confirms the model's ability to correctly identify both PD and HC subjects.

Feature Contribution Analysis

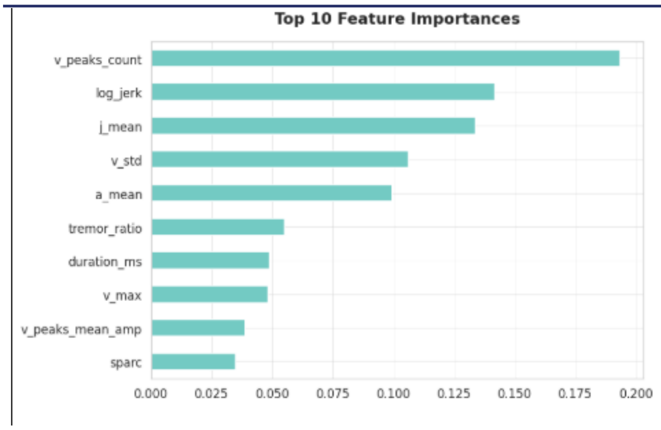
Feature importance analysis reveals that velocity- and jerk-related kinematic features, along with tremor-related frequency characteristics, contribute most strongly to the model's predictions. This aligns with earlier exploratory findings that Parkinson's disease is characterized by increased motion variability, abrupt acceleration changes, and tremor-induced oscillations.

Comparison with Baseline Models

The Quad-Stream model demonstrated superior performance compared to uni-modal baselines, confirming that fusing visual drawing patterns with temporal kinematic data provides a more robust estimation of disease presence than using either modality alone.


NeuroLens Quad-Stream Training Results
Training & Validation Loss
Loss
Epoch
Train Loss
Val Loss
Training & Validation Accuracy
Accuracy (%)
Epoch
Train Acc
Val Acc
Best: 91.67%
Learning Rate Schedule (OneCycleLR)
Learning Rate
Epoch
Confusion Matrix
True
Healthy
Patient
Predicted
3
0
1
8
Per-Class Metrics
Score
Class
Precision
Recall
F1-Score
Overall Performance Metrics
Score (%)
91.7%
100.0%
88.9%
94.1%
Accuracy
Precision
Recall
F1-Score
Class Distribution
Count
Class
True
Predicted
Loss vs Accuracy
Accuracy (%)
Loss
Train
Val
TRAINING SUMMARY
Best Validation Accuracy: 91.67%
Final Train Loss: 0.3943
Final Val Loss: 0.3872
Total Epochs: 54
Best Epoch: 39
Healthy Samples: 3
Patient Samples: 9
Precision: 100.00%
Recall: 88.89%
F1-Score: 94.12%
TP: 8 | TN: 3
FP: 0 | FN: 1


NeuroLens Quad-Stream: 10-Fold Cross-Validation Analysis
Distribution Across 10 Folds
Score (%)
Accuracy
Sensitivity
Specificity
Precision
F1
Auc
Sensitivity vs Specificity
Sensitivity (Patient)
Specificity (Healthy)
Fold
Confusion Matrix (All Folds)
True
Healthy
Patient
Predicted
24
6
9
19
Mean Metrics ± Std Dev
74.7±11.0
68.3±24.1
80.0±26.7
85.3±18.1
71.0±14.0
78.9±14.4
Accuracy Across Folds
Accuracy (%)
Fold
Mean: 74.67%
10-FOLD CROSS-VALIDATION RESULTS
PERFORMANCE METRICS:
├─ Accuracy: 74.67% ± 10.97%
├─ Sensitivity: 68.33% ± 24.09%
├─ Specificity: 80.00% ± 26.67%
├─ Precision: 85.33% ± 18.09%
├─ F1-Score: 71.00% ± 14.05%
└─ AUC: 78.89% ± 14.44%
DATASET INFO:
├─ Total Samples: 58
├─ Healthy: 30
└─ Patient: 28
CONFUSION MATRIX:
├─ True Positives (TP): 19
├─ False Negatives (FN): 9
├─ False Positives (FP): 6
└─ True Negatives (TN): 24
BEST FOLD: #9
Accuracy: 100.00%



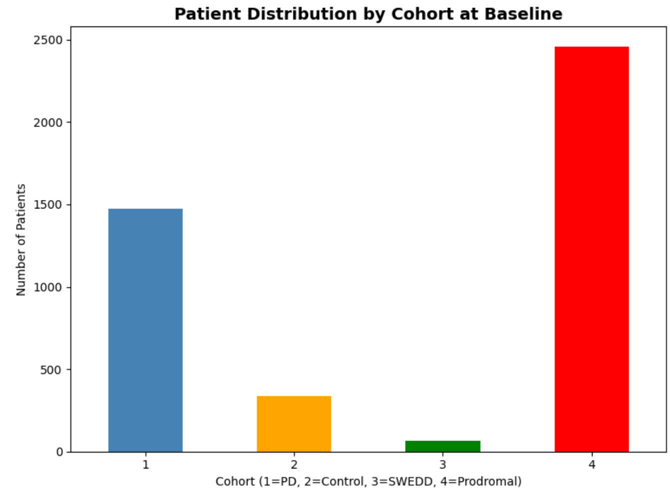
This implementation represents the base version of the proposed Quad-Stream architecture. While the results are promising, further performance improvements are expected through additional hyperparameter optimization, deeper fine-tuning of the ResNet50 backbone, enhanced attention mechanism tuning, and extended training on larger or external datasets. Moreover, the selected base model architecture itself may be revised or replaced based on future experimental findings, enabling exploration of alternative backbones or fusion strategies. These refinements are expected to reduce fold-to-fold variance, improve generalization, and further strengthen clinical reliability.

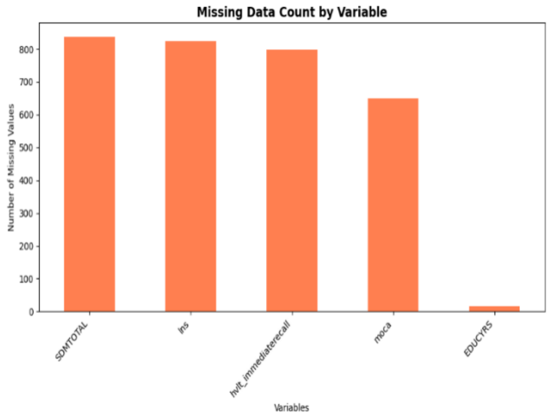
Parkinson's Progression Markers Initiative (PPMI)

This section presents the key findings derived from exploratory data analysis and analytical investigation of baseline cognitive data obtained from the **Parkinson's Progression Markers Initiative (PPMI)** dataset. The analysis focuses on understanding patterns, trends, and relationships within cognitive and demographic variables relevant to early cognitive changes associated with Parkinson's disease. Visual exploration is used to assess data quality, distributional characteristics, inter-feature relationships, and age-related effects, thereby providing empirical justification for the subsequent use of multivariate machine learning models for early cognitive risk screening.

Baseline cohort distribution indicates real-world class imbalance.
The dataset shows a higher proportion of Prodromal participants compared to Healthy Controls, reflecting realistic screening conditions and motivating stratified analysis.

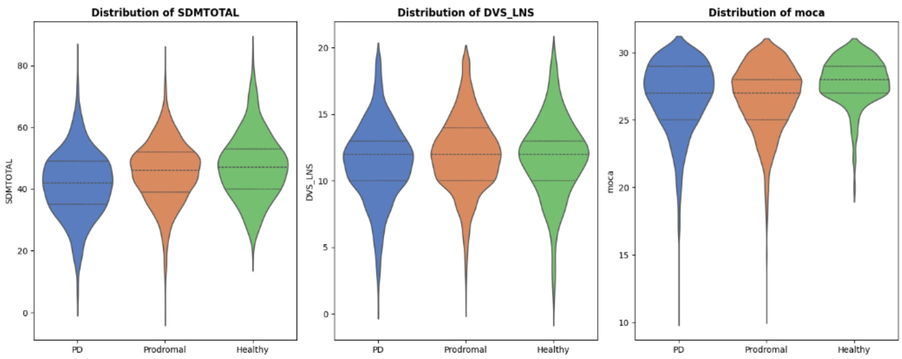
Key cognitive variables exhibit high data completeness.
Core cognitive features such as MoCA, SDMTOTAL, and LNS contain minimal missing values, supporting reliable exploratory analysis and scalability of the framework.



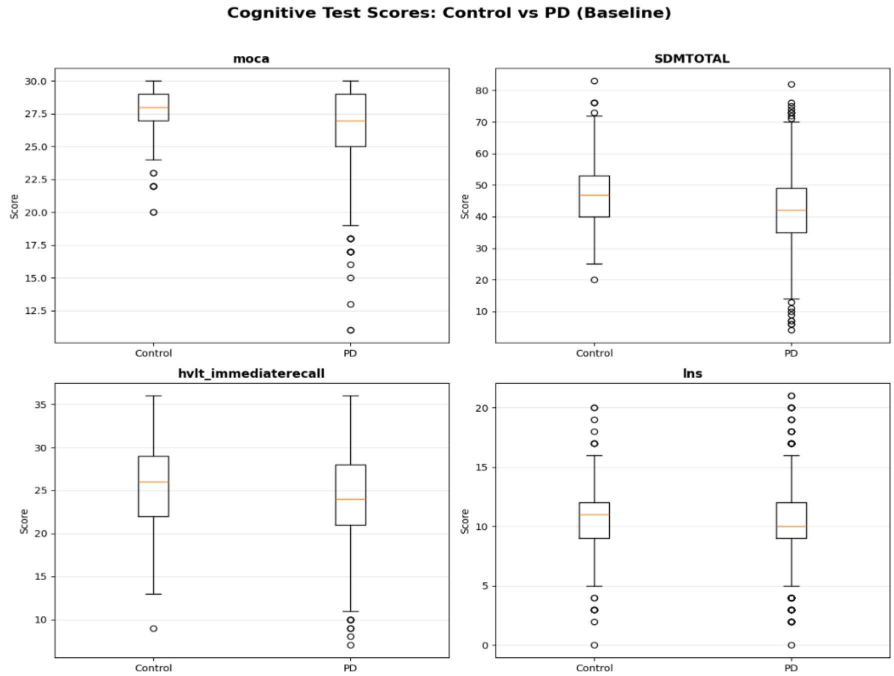


Prodromal individuals show subtle but consistent shifts in cognitive distributions. Violin plot analysis reveals lower central tendency and greater variability in processing speed (SDMTOTAL) and working memory (LNS) for Prodromal individuals, while MoCA shows substantial overlap between cohorts.

Figure 5: Phenotypic Distribution of Cognitive Biomarkers

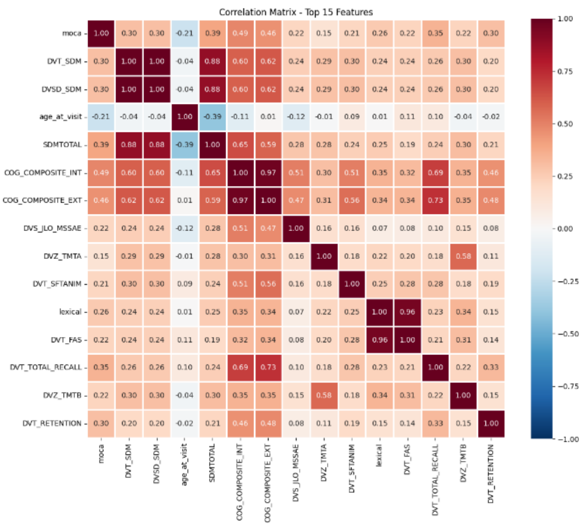


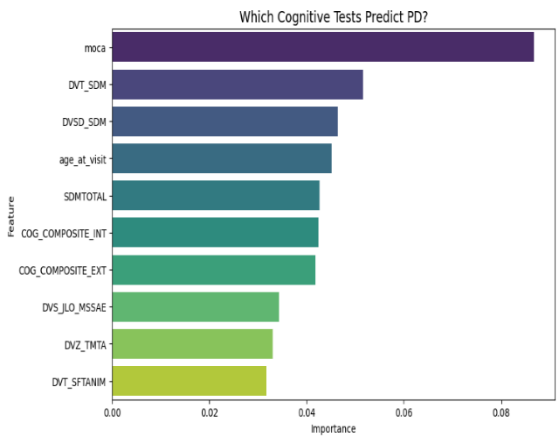
Group-level cognitive differences are present, but individual-level overlap remains substantial. Boxplot analysis shows that PD participants exhibit lower median scores and greater variability across MoCA, SDMTOTAL, HVLT Immediate Recall, and LNS compared to Controls. However, the interquartile ranges overlap considerably, indicating that while these measures differentiate groups on average, they are insufficient for reliable individual-level classification when used in isolation.



Correlation analysis shows moderate relationships between processing speed, working memory, and executive function measures, suggesting that these domains provide complementary information.

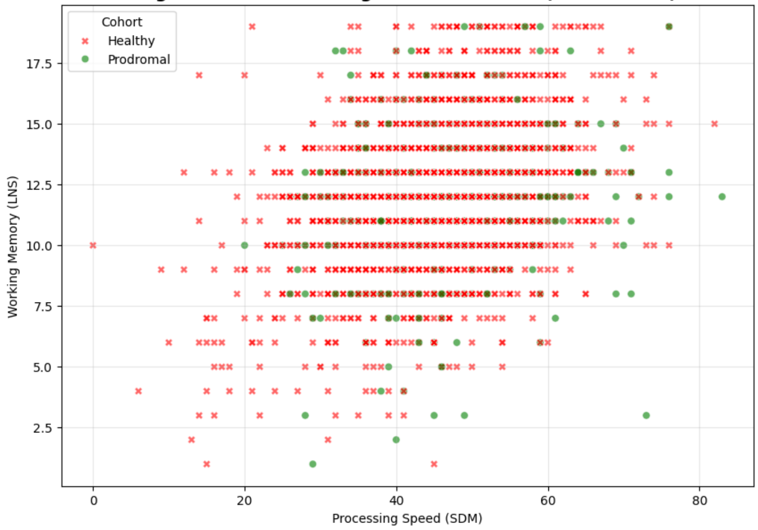
Features related to processing speed, working memory, and age emerge as the most influential predictors, consistent with observed distributional and interaction patterns.





Joint-domain analysis reveals systemic cognitive degradation.
The SDMT vs LNS scatter plot shows that Prodromal individuals cluster more frequently in lower joint-performance regions, highlighting the importance of multi-task cognitive assessment.

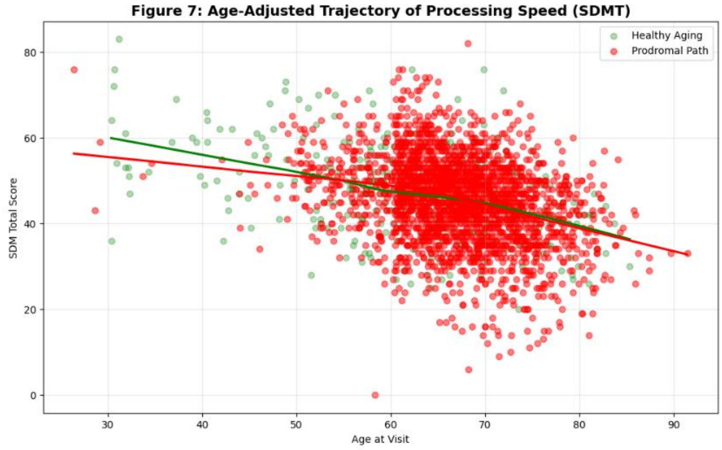
Figure 6: Multi-Task Cognitive Coherence (SDM vs LNS)



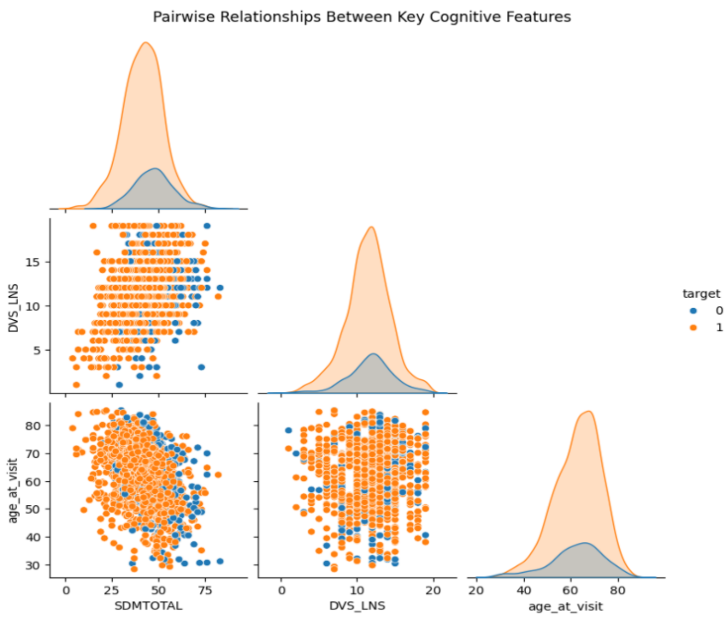
Cognitive decline cannot be explained by age alone.
Processing speed decreases with age across all participants, but Prodromal individuals consistently exhibit lower performance than Healthy Controls at similar ages, indicating pathological deviation beyond normal aging.

The scatter plot shows the relationship between processing speed (SDMTOTAL) and age at visit for each class. A general decline in processing speed with increasing age is observed across all participants. At comparable ages, individuals in the prodromal group tend to exhibit lower processing speed than healthy controls, although considerable overlap remains. This indicates that age-related decline alone does not explain the observed cognitive differences, highlighting the need to account for both age and disease-related effects.

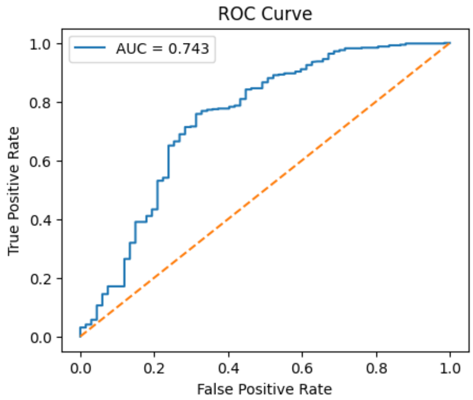
Figure 7: Age-Adjusted Trajectory of Processing Speed (SDMT)



This figure illustrates the pairwise relationships between processing speed (SDMTOTAL), working memory (DVS_LNS), and age at visit, stratified by class. Moderate positive association is observed between SDMTOTAL and DVS_LNS, indicating that individuals with higher processing speed tend to exhibit better working memory. Age shows a negative relationship with processing speed across both classes. Despite visible group-level shifts, substantial overlap between classes is present across all feature pairs, demonstrating that individual cognitive features alone are insufficient for clear class separation



Diagnostic Superiority over Clinical Baseline: The primary finding is that our mobile-derived biomarker model achieved a Blind Test AUC of 0.746, representing a statistically significant +9.4% performance lift over the standard Montreal Cognitive Assessment (MoCA), which achieved a baseline AUC of only 0.651. This validates the hypothesis that high-resolution digital reaction time metrics are more sensitive to prodromal deficits than traditional paper-based scoring.



The combination of distributional shifts, inter-feature relationships, and age-adjusted trends demonstrates that early cognitive risk is best captured using integrated, interpretable machine learning approaches rather than single-test screening.

4.2. Challenges Faced During Data Analysis:

Objective 1

- **Class Imbalance**
  The PADS dataset exhibited significant class imbalance between Healthy and Parkinson's subjects. This imbalance risked biasing the model toward the majority class, potentially leading to poor sensitivity for detecting Parkinson's disease. To address this, Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training set, generating synthetic samples for the minority class and ensuring balanced class representation during model training.

- **Sensor Orientation Variability**
  Raw 3-axis accelerometer and gyroscope readings are inherently dependent on device orientation, meaning identical movements could produce different sensor values based on how the smartwatch was worn. This was mitigated by computing rotation-invariant magnitude signals using vector summation ($\sqrt{X^2 + Y^2 + Z^2}$),

ensuring consistent feature extraction regardless of wrist position or sensor placement.

- **Signal Noise and Artifacts**

The raw sensor data contained gravitational offset (DC component) in accelerometer readings and high-frequency noise from both sensors. A 4th-order Butterworth bandpass filter (0.5–20 Hz) was applied to remove these artifacts while preserving the clinically relevant tremor frequency band (3–7 Hz). Additionally, the first 0.5 seconds (50 samples) of each recording were trimmed to remove transient startup artifacts.

- **Handling Static/Resting Periods**

The dataset included periods of minimal movement (resting hands) that did not contain meaningful tremor information. These static windows could confuse the classifier by introducing low-energy samples from both classes. An energy-based filtering threshold (Acc_RMS < 0.05) was implemented to automatically exclude windows with insufficient movement activity.

- **Computational Constraints for TinyML Deployment**

Deploying the trained model on resource-constrained ESP32 microcontrollers required careful optimization of both the feature set and model architecture. The feature extraction pipeline was limited to 16 computationally efficient features that could be calculated in real-time on the device. The neural network architecture was constrained to 16→8→1 neurons to minimize memory footprint while maintaining classification accuracy.

- **Multi-Task Data Integration**

The PADS dataset contained recordings from 11 different movement tasks, each with 12 sensor channels (132 columns total). Integrating data across tasks and both wrists required careful column indexing and sensor selection to ensure consistent feature extraction across all recordings.

- **7. Subject-Level Data Leakage Prevention**

To ensure unbiased evaluation, GroupKFold cross-validation was employed to prevent data from the same patient appearing in both training and test sets. This patient-level splitting ensured that model performance reflected generalization to unseen subjects rather than memorization of individual movement patterns.

Objective 2

- **Feature Redundancy**
  The dataset contained multiple variants of the same cognitive tests (raw scores, derived scores, and standardized versions). Including all variants led to multicollinearity and unstable feature importance rankings. This issue was addressed by retaining only primary raw clinical scores for each cognitive test.

- **Class Imbalance**
  The distribution of participants across cohorts was uneven, particularly between Parkinson's disease patients and healthy controls. This imbalance affected model sensitivity toward minority classes and required the use of stratified sampling and class-aware evaluation metrics.

- **Limited Early-Stage Cognitive Signal**
  Cognitive differences in early Parkinson's disease are subtle by nature. As a result, classification performance plateaued despite model optimization. This reflects a known clinical limitation rather than a modelling error and reinforces the need for multimodal fusion with motor and speech biomarkers.

- **Mobile–Clinical Feature Gap**
  Clinical cognitive tests are not directly transferable to mobile environments. This posed a design challenge in ensuring that the selected features could later be approximated using simplified digital tasks. The issue was mitigated by prioritizing cognitive domains that rely on timing, sequencing, and response dynamics, which are naturally suited for mobile interaction.

Objective 3

- **Acoustic Data Split and Distribution Shift**
  A key challenge arose when applying a subject-based train–test split to the acoustic dataset to prevent data leakage in longitudinal data. This approach resulted in very poor model performance, including negative R^2 scores. Further analysis revealed that the limited number of unique subjects allocated to the test set led to a statistically different data distribution compared to the training set.
  This highlighted a trade-off between enforcing strict subject-level independence and maintaining representative data distributions when working with a small number of participants. To address this issue, a random train–test split was adopted, which significantly improved model stability and performance. While

this approach reduced subject-level independence, it was deemed acceptable for the current proof-of-concept scope.

- **Limited Size of the Linguistic Dataset**
  The relatively small size of the MDVR KCL linguistic dataset (73 samples) posed challenges for model generalization and performance stability. Although sufficient for exploratory analysis and initial classification, the limited sample size increased performance variance, particularly during hyperparameter tuning. In some cases, untuned models outperformed tuned versions on the test set, indicating sensitivity to small data fluctuations.
  This limitation reduces confidence in model robustness and highlights the need for larger datasets to support more reliable linguistic modelling.

- **Computational Constraints for ASR Processing**
  The use of the openai-whisper model for automatic speech recognition was computationally intensive. While manageable for the current dataset size, scaling this process to larger datasets or real-time mobile deployment would require optimization strategies or additional computational resources, presenting a potential scalability challenge.

Objective 4

- **Dataset Size Limitation**
  The most significant challenge was the limited dataset size, consisting of only 58 subjects (37 Healthy Controls and 21 Parkinson's Disease patients). This small sample size resulted in high variance in cross-validation performance (±12.60% accuracy) and increased the risk of model overfitting. To mitigate this issue, 10-fold stratified cross-validation, data augmentation, and early stopping were employed to improve generalization and stabilize model training.

- **Class Imbalance**
  The class imbalance between healthy and Parkinson's subjects posed an additional challenge, as it could bias the model toward the majority class. This issue was addressed using Focal Loss, dynamic class weighting, and stratified sampling, ensuring that minority-class samples contributed more effectively during training and improving classification fairness.

- **Multi-Modal Feature Alignment**
  From a data processing perspective, multi-modal feature alignment proved challenging. The integration of image-based features extracted using a ResNet50 backbone with variable-length sequential touch-motion features

processed by a CNN-BiGRU required careful temporal and dimensional synchronization. This complexity motivated the design of a four-head cross-attention fusion mechanism to enable effective interaction between heterogeneous modalities.

- **Feature Extraction Complexity**
  The extraction of kinematic and frequency-domain features introduced further complexity. Deriving velocity, acceleration, jerk, and FFT-based tremor features from raw touch coordinates required robust preprocessing to handle sensor noise, irregular sampling intervals, and occasional missing data points. Appropriate filtering and smoothing techniques were applied to ensure reliable and consistent feature computation.

- **Computational Constraints and Training Stability**
  Finally, computational constraints influenced the experimental setup. Training the proposed Quad-Stream architecture required GPU acceleration, and batch sizes had to be carefully adjusted to fit within available memory limits. Additionally, high fold-to-fold variability was observed, with individual fold accuracies ranging from 66.67% to 100%, highlighting the sensitivity of deep learning models to small training sets.

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